

Strategically Controlling Worldviews*

Cuimin Ba Danil Dmitriev Ziqi Hang Freddie Papazyan
Johns Hopkins University University of Georgia Texas Tech University Texas Tech University

October 1, 2026

[Click here to see the most recent version](#)

Abstract

This paper studies persuasive behavior when the sender can both control the information the receiver observes (the *message*) and propose a model of its interpretation (the *narrative*). We show that the sender can manipulate the receiver into always taking her preferred action, even when they start with a correctly specified model of her strategic behavior, update by Bayes' rule given their model, and are skeptical of her narrative. The key mechanism highlights a strong complementarity between strategic communication of information and narratives, which allows the sender to strictly outperform Bayesian persuasion without commitment power. We fully characterize the sender-optimal equilibrium for a broad class of information technologies. Softer information lowers the bar for full manipulation, while harder information expands the set of environments where any manipulation is possible. The results provide a formal foundation for understanding the widespread success of disinformation.

Keywords: Disinformation, narratives, cheap talk, disclosure, persuasion

JEL Codes: C72, D72, D82, D83, D90

*First version: March 23, 2026. Ba: bacuimin@gmail.com. Dmitriev: danil.dmitriev@uga.edu. Hang: victoria.hang@ttu.edu. Papazyan: freddie.papazyan@ttu.edu. The authors would like to thank Alejandro Abarca, Dilip Abreu, Arjada Bardhi, B. Douglas Bernheim, Aislinn Bohren, Henrique Castro-Pires, Carlo Cusumano, Sulagna Dasgupta, Rahul Deb, Théo Durandard, Kfir Eliaz, Hülya Eraslan, Mira Frick, Tristan Gagnon-Bartsch, Duarte Gonçalves, Kevin He, Ryota Iijima, Navin Kartik, Botond Köszegi, Remy Levin, Jonathan Libgober, Xiao Lin, George Mailath, Franz Ostrizek, Harry Pei, Jacopo Prego, Daniel Rappoport, Anne-Katrin Roesler, Denis Shishkin, Joel Sobel, Ran Spiegler, Natacha Spitzer, Philipp Strack, Jean Tirole, Yu Fu Wong, and Beixi Zhou for insightful discussions. We are also grateful for the helpful comments from conference audiences at SITE, Bonn, Cowles, SAET, Stony Brook, SWET, WEAI, and the Texas Economic Theory Camp, and from seminar attendees at Texas Tech, Pitt, Penn, LSE, UBC, UNC Chapel Hill, Duke, and the Toulouse School of Economics.

Certainly anyone who can make you believe
absurdities can make you commit injustices.
(*Certainement qui est en droit de vous rendre
absurde, est en droit de vous rendre injuste.*)

Voltaire, *Questions sur les Miracles*, 1765

1 Introduction

We are living in a troubling chapter of the information age, where it is increasingly hard to know who or what to trust. With abundant data at their disposal, strategic actors systematically manipulate perceptions and beliefs, persuading others to act against their own interests. Yet the widespread success of such manipulation represents a puzzle for standard economic theory: if individuals rationally evaluate information and update beliefs using Bayes’ rule, unreliable sources should lose credibility and play little role in decision making (Akerlof and Shiller, 2015; Stiglitz and Kosenko, 2024, p. 64).

The key to resolving this puzzle, we suggest, lies in understanding what persuaders actually communicate. Consider an opinion leader posting data on social media alongside a misleading interpretation, a finance influencer recommending a failing stock while framing it as an undervalued opportunity, a politician releasing favorable employment numbers while attributing them to their policies, or a university inflating grades while presenting A’s as true signals of excellence. In each case, the persuader communicates not only information about the state of the world but also a “narrative” about how to interpret that information. Although such joint use is pervasive in practice, the economic literature has largely studied strategic communication of information (Sobel, 2020b) and persuasion through narratives (Schwartzstein and Sunderam, 2021) separately.

We develop a unifying framework that combines both and show that systematic manipulation is robustly possible, thereby providing a formal foundation for disinformation. We study a classic setting where a sender seeks to convince a receiver to take a risky action over a safe one. The sender privately observes a state that determines the receiver’s payoff from the risky action. She then communicates a message using her information technology but, as in cheap talk, cannot commit to a messaging strategy in advance. Before the state is realized, the sender can propose a narrative: a model describing how she intends to communicate and thus how messages should be interpreted. To stack the environment against the sender, we consider a strategically sophisticated receiver who is endowed with a correctly specified model of the sender’s incentives and information technology under which he correctly anticipates the sender’s messaging strategy in equilibrium and updates beliefs

using Bayes’ rule.¹ He is not, however, dogmatically certain that his initial model is correct. He therefore considers the proposed alternative while remaining skeptical of it. After observing the message, he compares how well his initial model and the proposed model explain its occurrence and switches to the proposed model if the Bayes factor—the likelihood ratio of the proposed model to the initial model given the observed message—reaches a fixed threshold, with a higher threshold representing greater skepticism (Ba, 2026).

To fix ideas, consider a finance influencer who privately knows whether a stock is worth buying (the state), but always wants to convince her followers to buy since she holds a position in it. Before trading begins, the influencer can present herself as a particular type of communicator—for instance, claiming to be an “honest” analyst who only recommends stocks she genuinely believes in (the narrative)—and chooses what recommendations to share with her followers (the message). Her followers initially understand her strategic incentives and are skeptical of her claimed honesty, but are willing to update their interpretation when her messages seem sufficiently consistent with the narrative. When the stock is not worth buying, persuading her followers to purchase it is precisely the manipulation at the heart of a “pump-and-dump scheme.”²

We characterize when the sender can *fully manipulate* the receiver by inducing the risky action with probability one, *partially manipulate* him by inducing it more often than through strategic messaging alone, and when narratives provide no additional benefit. To study how persuasiveness depends on information technology, we analyze a full spectrum of cases ranging from cheap talk to mandatory disclosure. We find that narratives dramatically change the usual technology rankings in pure messaging models: softer information lowers the bar for full manipulation, while harder information expands the set of environments where any manipulation is possible.

We begin with the cheap talk benchmark, where the sender can send any message. While babbling is the unique equilibrium in standard cheap talk, we show that with narratives the sender either fully manipulates the receiver or cannot manipulate at all, with the outcome determined by a simple demand-budget comparison that depends on the state prior and the

¹This rules out explanations of disinformation based on receivers misunderstanding from the start the sender’s incentives or how her messages should be interpreted. An extensive literature in psychology and marketing documents that listeners often understand when and how a speaker may be trying to influence their actions and interpret messages in light of the speaker’s incentives (Friestad and Wright, 1994; Isaac and Calder, 2025).

²In a “pump-and-dump scheme,” manipulators use misleading positive claims to artificially inflate (“pump”) an asset’s price before selling it (“dump”). This is widely observed in microcap and penny stock markets (Leuz et al., 2026) and more recently in cryptocurrency markets (Dhawan and Putniņš, 2023) alongside a closely related, newer form of fraud known as a “rug pull” (Agarwal et al., 2023).

receiver’s skepticism ([Theorem 1](#)). Full manipulation is achieved through a *confidence trick*: despite having no commitment power, the sender claims to communicate as a “Bayesian persuader,” sending a low message in bad states and a high message in a pool of relatively good states,³ but actually reverses the mapping by sending the high message in bad states and an extra-high message in good states. If the standard high message is sufficiently more likely under the proposed narrative than under the sender’s true strategy, the receiver switches models and takes the risky action. On the other hand, the extra-high message deliberately contradicts the sender’s narrative but credibly conveys that the state is favorable, so the receiver again takes the risky action. Full manipulation is feasible if and only if the *demand* for narratives (i.e., the probability mass of bad states) can be covered by the *budget* (i.e., the probability mass of good states) at a *price* equal to the receiver’s switching threshold. Central to this result is the complementarity between strategic messaging and narratives: the narrative covers bad states by flipping the interpretation of losing messages, freeing the sender to use strategic messaging at full capacity as in Bayesian persuasion (henceforth BP).

Because this condition depends only on tail probability mass rather than the full distribution of states, stark welfare implications follow ([Corollaries 1 and 2](#)). Full manipulation remains feasible even when the risky action offers a vanishing upside and arbitrarily severe downside compared to the safe action, resulting in extreme welfare loss for the receiver. Moreover, a more favorable state distribution can make the receiver discontinuously worse off, since it enables manipulation that was previously infeasible.

In the finance example, the confidence trick takes the form of recommendation inflation: the influencer claims to recommend only relatively good stocks as if she had commitment power but actually issues a “strong buy” for them and a “buy” for bad stocks.⁴ Her followers observing “buy” switch models and buy, believing it signals a genuine endorsement; those seeing “strong buy” correctly infer good news and also buy. The influencer thus fully manipulates all her followers, and can do so even when the stock has arbitrarily negative expected return. When good stocks are too rare, a “buy” recommendation would be too unlikely to come from the type of influencer she claims to be, the narrative becomes implausible, and manipulation fails entirely.

We next generalize the analysis to a broad class of *monotone interval information tech-*

³The pool of good states is chosen as in optimal Bayesian persuasion so that, under the claimed strategy, the high message just makes the receiver willing to take the risky action.

⁴[Malmendier and Shanthikumar \(2007\)](#) find that small investors follow analysts’ inflated recommendations literally (buying after both “buy” and “strong buy” recommendations) and discuss information cost and investor naïveté as possible explanations. Our mechanism can generate this behavior even among strategically sophisticated and skeptical receivers.

nologies, in which each type’s feasible messages form an interval that shifts upward with the true state, with cheap talk as a special case. We characterize the solution to the constrained Bayesian persuasion problem (Proposition 1) and the sender-optimal equilibrium without narratives (Proposition 2) for this full class, results which, to the best of our knowledge, are novel in the disclosure literature. With narratives, the demand-budget logic extends: the sender fully manipulates whenever her narrative budget covers the residual demand left unmet by strategic messaging, where residual demand is now shaped by the technology (Theorem 2). While harder to obtain, full manipulation remains feasible with the hardest possible evidence, because narratives allow the sender to reframe how bad evidence is interpreted.

Consider the finance example again, but now suppose that regulation requires the influencer to truthfully disclose past returns, which are perfectly positively correlated with future returns. Followers begin with a correctly specified model that incorporates this requirement and supports a momentum strategy, though they are not completely certain of it. Since evidence cannot be altered, the confidence trick works by reversing its interpretation rather than concealing or inflating it. The influencer promotes a contrarian narrative, claiming that poor past returns signal an undervalued opportunity with strong future performance, while strong past returns signal an overvalued stock headed for a decline. If stocks with strong future returns are sufficiently common, poor past returns are sufficiently more likely under the contrarian narrative to persuade investors to adopt it and buy. Strong past returns instead favor the initial momentum model, so investors retain it and also buy.

When full manipulation is infeasible, message constraints give rise to partial manipulation, a possibility absent under cheap talk (Theorem 3). Two regimes emerge depending on whether the technology can credibly exclude bad types to stabilize the losing region. If so, the sender combines strategic messaging and narrative persuasion in equilibrium, and complementarity again drives her payoff strictly above the BP benchmark. If not, the two instruments decouple: the sender relies on narrative persuasion alone, and need not surpass her BP payoff.

These characterizations reveal a key tradeoff in how information hardness affects persuasion. On the one hand, conventional wisdom in pure communication models holds that harder information allows the sender to credibly convey more in equilibrium. On the other hand, harder information makes it harder to misrepresent information, suggesting it should limit manipulation. Our results show that both forces manifest when narratives are available, but with distinct implications: softer information facilitates full manipulation by expanding messaging *flexibility* and reducing the demand for narratives, while harder

information expands the scope for any manipulation by, paradoxically, restoring the *credibility* that cheap talk lacks. One technology that best resolves this tradeoff allows the sender to understate but not overstate the state: it minimizes the narrative demand while ensuring bad types can never mimic good ones, making it *maximally persuasive* across environments (Theorem 4).

The remainder of this section reviews related literature. Section 2 introduces the framework. Section 3 analyzes the cheap talk benchmark and Section 4 extends the analysis to general information technologies. Section 5 studies how information technology affects persuasiveness. Section 6 outlines several extensions such as Bayesian model averaging and multiple narratives, and Section 7 briefly discusses further applications and policy implications. All proofs of the results in the main text can be found in Appendix A. The Supplemental Appendix includes additional results and proofs.

Related Literature Our framework nests several models of communication and persuasion. In strategic communication (see Sobel, 2020b, for a survey), cheap talk is uninformative when interests are sufficiently misaligned, while partially verifiable information sustains informative equilibria by restoring credibility at the cost of flexibility (Hart et al., 2017; Lizzeri et al., 2026). Bayesian persuasion bypasses this tension with commitment, bounding the sender’s payoffs in pure messaging games (Kamenica and Gentzkow, 2011).⁵ Our sender additionally proposes narratives that change how messages are interpreted, inducing posteriors that are not Bayes-plausible and “completely inaccurate” by ruling out the true state (Sobel, 2020a). Combining narratives and strategic messaging thus allows the sender to strictly surpass Bayesian persuasion without commitment power.⁶

This paper contributes to the literatures on model misspecification and narrative persuasion. A growing literature studies belief convergence and persistence under exogenous misspecified models in individual and social learning (Heidhues et al., 2018; Bohren and Hauser, 2021; Frick et al., 2023). Ba (2026) studies the long-run persistence of misspecified models under the same Bayes factor switching rule we adopt; we instead study a strategic setting in which the sender endogenously proposes the alternative model. On the other

⁵Lipnowski et al. (2022) study a sender who announces a messaging strategy but loses commitment power with some probability, allowing her to deviate afterwards. Our sender also claims she will follow one strategy while choosing another, but she is never bound by her announcement.

⁶This paper is also connected to the literature on persuasion with non-standard receivers (e.g., de Clippel and Zhang, 2022). Galperti (2019) studies Bayesian persuasion with exogenously given worldviews; Ko (2025) studies a related setting in which the receiver’s default model treats the experiment as uninformative. In our framework, the receiver can be fully manipulated despite starting with a correctly specified model.

hand, work on narrative persuasion often endogenizes models while holding the information source fixed. [Schwartzstein and Sunderam \(2021\)](#) consider a sender who proposes a model of interpretation after observing the signal realization, allowing her to tailor narratives to the evidence.⁷ [Bauch and Foerster \(2024\)](#) study cheap-talk communication of narratives themselves that signal the sender’s private information about the data-generating process. In our framework, narratives are proposed ex ante, shutting down any signaling effect. [Aina \(2025\)](#) shares this timing and studies how multiple competing narratives can make all signal realizations favorable to the sender; our framework differs in allowing the sender to jointly control both the narrative and the messaging strategy, expanding the scope for manipulation.

Several recent papers study information and narratives in related settings. [Eliaz et al. \(2021\)](#) study a sender who pairs information with a truthful but coarse description of her strategy that the receiver uses directly; in contrast, our narratives need not match the sender’s true strategy and the receiver begins with a correctly specified model. [Eliaz and Spiegler \(2026\)](#) study a news outlet that commits to a Blackwell experiment and a causal narrative specifying whether the consumer’s action and/or the state cause outcomes. Our sender lacks commitment, and narratives are proposed messaging strategies drawn from a rich space. Both papers identify complementarity between information and narratives, but through different mechanisms.⁸ [Ichihashi and Meng \(2021\)](#) also consider a sender who commits to a Blackwell experiment and proposes an interpretation ex post; our sender proposes ex ante without commitment. [Jain \(2023\)](#) studies a narrator who interprets information from a benevolent information provider; our self-interested sender supplies both information and its interpretation, though she may equivalently represent two parties with aligned incentives controlling them separately. Beyond these differences, our paper is the first to jointly study strategic communication and narrative persuasion across a general class of information technologies.

⁷Relatedly, [Schwartzstein and Sunderam \(2024\)](#) study the exchange of models in social interactions.

⁸In [Eliaz and Spiegler \(2026\)](#) the outlet maximizes engagement, measured by consumers’ anticipatory utility given the narrative; our sender seeks to induce a particular action, and the receiver adopts the narrative only when the message provides sufficient support. In their main application, optimistic signals induce action in unfavorable states, and an empowering narrative attributes the resulting success to consumers’ actions, thereby raising anticipatory utility. In ours, narratives make otherwise losing messages persuasive and prevent strategic messaging from unraveling.

2 Framework

2.1 Model

Preliminaries The players are a sender (S, she) and receiver (R, he). The receiver chooses an action $a \in \mathcal{A} = \{0, 1\}$. The safe action ($a = 0$) yields a known payoff $\omega_0 \in \mathbb{R}$, and the risky action ($a = 1$) yields an uncertain payoff equal to the state $\omega \in \Omega \subset \mathbb{R}$. The state ω is drawn by nature from a common prior $F \in \Delta\Omega$ with a finite mean and privately observed by the sender, interpreted as her *type*. We assume that either Ω is finite or it is a possibly unbounded closed interval and F admits a continuous, strictly positive density f . While the receiver's payoff depends on the chosen action and the realized state, the sender only cares about whether the risky action is taken. Specifically, $U_R(a, \omega) := (1 - a)\omega_0 + a\omega$ and $U_S(a) := a$. We focus on the non-trivial case where $\mathbb{E}[\omega] < \omega_0$ and $F(\omega \geq \omega_0) > 0$ so that the receiver strictly prefers the safe action absent additional information but the risky action can be optimal in some state.

Information technology After learning the state, the sender communicates a message m from a set $M(\omega) \subseteq M$, where M is the message space and $M(\omega)$ is the feasible message set in state ω . We assume M contains Ω and at least two additional messages so that M is rich enough to communicate any state or action recommendation. A *messaging strategy* is a mapping $\sigma : \Omega \rightarrow \Delta M$ from states to distributions over messages. Let $\bar{\Sigma}$ denote the set of all such mappings. Feasibility requires that the sender only sends messages available in the realized state, so the space of *feasible messaging strategies* is

$$\Sigma := \{\sigma : \text{supp}(\sigma(\cdot|\omega)) \subseteq M(\omega), \forall \omega \in \Omega\}. \quad (1)$$

The sender cannot commit to a messaging strategy in advance, but chooses a feasible message given the state. The sender's *information technology* $\mathcal{M} := \{M(\omega)\}_{\omega \in \Omega}$ summarizes the constraints on what she can say as a function of the state, capturing the extent to which her information is hard or soft. We accommodate a broad spectrum of information technologies, spanning from unrestricted messaging ("cheap talk"), where $M(\omega) = M$ for all ω , to mandatory disclosure, where $M(\omega) = \{\omega\}$ for all ω . Other choices of $M(\omega)$ capture intermediate restrictions on what the sender can report.

Model uncertainty The receiver initially holds a correctly specified model of the decision environment, including the sender's incentives and information technology constraints. As is standard in strategic communication, retaining this model implies correctly

anticipating the sender's equilibrium strategy, including her messaging strategy σ (where $\sigma = \sigma^*$ in equilibrium, as formally stated in [Definition 1](#)). However, he is not dogmatically certain of this model. Before the state is realized, the sender can propose a narrative specifying an alternative messaging strategy $\hat{\sigma} \in \bar{\Sigma}$, seeding an alternative model in the receiver's mind.^{9,10} Since the proposal itself is unrestricted cheap talk, it need not coincide with her actual messaging strategy and may violate the constraints imposed by $\mathcal{M}$. The receiver now considers both models possible but has no well-defined Bayesian prior over them. Instead, he adopts the proposed model when it is sufficiently consistent with the observed message relative to the initial model, following a rule based on *Bayes factors* ([Schwartzstein and Sunderam, 2021](#); [Ba, 2026](#)).

Formally, let P_σ and $P_{\hat{\sigma}}$ denote the unconditional distributions of messages induced by σ and $\hat{\sigma}$. For any measurable set of messages $\tilde{M} \subset M$,

$$P_\sigma(\tilde{M}) := \int_{\Omega} \sigma(\tilde{M}|\omega) dF(\omega) \quad \text{and} \quad P_{\hat{\sigma}}(\tilde{M}) := \int_{\Omega} \hat{\sigma}(\tilde{M}|\omega) dF(\omega). \quad (2)$$

Upon observing m , the receiver computes the Bayes factor $\lambda : M \rightarrow [0, +\infty]$,

$$\lambda(m) \equiv \lambda(m|\sigma, \hat{\sigma}) := \frac{dP_{\hat{\sigma}}/d\nu_{\sigma, \hat{\sigma}}}{dP_\sigma/d\nu_{\sigma, \hat{\sigma}}}(m) \quad \text{where} \quad \nu_{\sigma, \hat{\sigma}} := P_\sigma + P_{\hat{\sigma}}. \quad (3)$$

When message distributions are discrete, this reduces to $\lambda(m) = P_{\hat{\sigma}}(m)/P_\sigma(m)$.¹¹ Intuitively, $\lambda(m)$ compares how well the two models explain the observation of m ; it is zero when m can arise only under σ , and infinite in the reverse case.

The receiver switches to the proposed model if and only if $\lambda(m) \geq \alpha$ for a fixed *switching threshold* $\alpha > 0$. After observing m , the receiver believes the sender is using messaging strategy $\sigma_R \in \{\sigma, \hat{\sigma}\}$, where

$$\sigma_R \equiv \sigma_R(m|\sigma, \hat{\sigma}) := \begin{cases} \hat{\sigma} & \text{if } \lambda(m|\sigma, \hat{\sigma}) \geq \alpha \\ \sigma & \text{if } \lambda(m|\sigma, \hat{\sigma}) < \alpha \end{cases}. \quad (4)$$

To convince the receiver to switch, the message must be at least α times as likely to be sent under $\hat{\sigma}$ as under σ , so α measures the receiver's skepticism toward the narrative. The

⁹We assume the narrative is proposed before the state is realized to shut down any signaling effect. One interpretation of this restriction is that the sender faces a heterogeneous population of receivers with idiosyncratic states and can only broadcast a single model to all of them. Moreover, our full-manipulation equilibria attain her first best in every state with one narrative, so state-contingent narratives are unnecessary.

¹⁰In [Section 6](#) we consider multiple models proposed before the state is realized.

¹¹Using $\nu_{\sigma, \hat{\sigma}}$ as the dominating measure accommodates various types of distributions. When both message distributions admit densities $p_{\hat{\sigma}}$ and p_σ with respect to a common reference measure, the definition reduces to $\lambda(m) = p_{\hat{\sigma}}(m)/p_\sigma(m)$. Values of the densities at points where they are not uniquely determined are specified as part of the equilibrium construction.

narratives literature commonly takes $\alpha = 1$ (Schwartzstein and Sunderam, 2021). If $\alpha < 1$, the receiver is inclined to trust the sender and switches even when m is less likely under $\hat{\sigma}$. Larger values make him more skeptical; if $\alpha = \infty$, he never switches on equilibrium path, reducing the game to standard strategic communication.

Belief updating Given the adopted σ_R and message m , the receiver forms a posterior belief $\pi_{\sigma_R}(\cdot|m) \in \Delta\Omega$ using Bayes' rule. For any measurable $\tilde{\Omega} \subset \Omega$,

$$\pi_{\sigma_R}(\tilde{\Omega}|m) = \frac{\int_{\tilde{\Omega}} \sigma_R(dm|\tilde{\omega})dF(\tilde{\omega})}{\int_{\Omega} \sigma_R(dm|\omega)dF(\omega)}, \quad (5)$$

whenever the denominator is positive.¹² The receiver then chooses an optimal action to maximize his expected payoff given posterior π_{σ_R} .

Timing The game proceeds as follows.

1. The sender proposes a model that specifies a messaging strategy $\hat{\sigma} \in \bar{\Sigma}$.
2. Nature draws the state $\omega \sim F$, which is privately observed by the sender.
3. The sender sends a message m according to messaging strategy $\sigma \in \Sigma$.
4. The receiver decides whether to switch models, thereby selecting the messaging strategy $\sigma_R \in \{\sigma, \hat{\sigma}\}$ that he believes the sender is using.
5. The receiver forms a posterior belief $\pi_{\sigma_R}(\cdot|m) \in \Delta\Omega$.
6. The receiver chooses an action $a \in \mathcal{A}$, and payoffs $U_R(a, \omega)$ and $U_S(a)$ are realized.

Strategies and equilibrium The sender's strategy consists of a *narrative strategy* $\hat{\sigma} \in \bar{\Sigma}$ and a *messaging strategy* $\sigma : \bar{\Sigma} \rightarrow \Sigma$, where $\sigma(\hat{\sigma})$ specifies her messaging strategy following proposal $\hat{\sigma}$. Under the initial model, the receiver correctly anticipates her full equilibrium strategy $(\hat{\sigma}^*, \sigma^*(\cdot))$. We often simply write σ^* for the on-path messaging strategy $\sigma^*(\hat{\sigma}^*)$. Let $A : \Delta\Omega \rightarrow \Delta\mathcal{A}$ denote the receiver's action strategy. Our equilibrium concept, which we call weak *Narrative PBE* (henceforth *equilibrium*), adapts weak Perfect Bayesian equilibrium to accommodate model switching. For convenience, our constructions use pessimistic beliefs for off-path narratives and messages. We discuss equilibrium refinements in Supplemental Appendix B.1.

Definition 1. A weak *Narrative PBE* is a tuple $(\hat{\sigma}^*, \sigma^*, A^*, \pi^*, \sigma_R^*)$ that satisfies:

¹²For continuously distributed messages, Bayes' rule determines posteriors only almost everywhere. We additionally require that, at every on-path message m , the posterior assign probability one to $\{\omega \in \Omega : m \in \text{supp}(\sigma_R(\cdot|\omega))\}$ whenever this set is nonempty.

(i) Optimality of the narrative strategy:

$$\hat{\sigma}^* \in \arg \max_{\hat{\sigma} \in \bar{\Sigma}} \mathbb{E}_{\omega \sim F, m \sim \sigma^*(\cdot|\omega, \hat{\sigma})} \left[U_S \left(A^* \left(\pi_{\sigma_R^*(m|\sigma^*(\hat{\sigma}), \hat{\sigma})}^*(\cdot|m) \right) \right) \right]. \quad (6)$$

(ii) Optimality of the messaging strategy: for every ω and $\hat{\sigma}$,

$$\sigma^*(\cdot|\omega, \hat{\sigma}) \in \arg \max_{\sigma \in \Sigma} \mathbb{E}_{m \sim \sigma(\cdot|\omega, \hat{\sigma})} \left[U_S \left(A^* \left(\pi_{\sigma_R^*(m|\sigma^*(\hat{\sigma}), \hat{\sigma})}^*(\cdot|m) \right) \right) \right]. \quad (7)$$

(iii) $\sigma_R^*(m|\sigma, \hat{\sigma})$ is determined by Eq. (4) whenever possible on the equilibrium path.¹³

(iv) $\pi_{\sigma_R}(\cdot|m)$ is updated by Bayes' rule whenever possible on equilibrium path, and, at every on-path message m , assigns probability one to $\{\omega \in \Omega : m \in \text{supp}(\sigma_R(\cdot|\omega))\}$ whenever this set is nonempty.

(v) $A^*(\pi) \in \arg \max_{a \in \mathcal{A}} \mathbb{E}_{\omega \sim \pi} [U_R(a, \omega)]$ for all $\pi \in \Delta\Omega$.

Intuitively, in equilibrium (i) the sender proposes a model to maximize her expected payoff anticipating her own equilibrium messaging strategy and the receiver's correct conjecture about it; (ii) her messaging strategy maximizes her expected payoff given any state and proposed model; (iii)–(iv) the receiver adopts models via the Bayes factor rule and forms posterior beliefs using Bayes' rule whenever possible on equilibrium path; and (v) the receiver chooses actions to maximize his expected payoff.

Objects of interest We focus on the *sender-optimal equilibrium*, i.e., the equilibrium that maximizes the sender's expected payoff. This provides a natural selection criterion under multiplicity and identifies the maximum gain from jointly choosing the narrative and messaging strategy. We then assume without loss that the receiver breaks ties in favor of the risky action. We ask how much narratives improve the sender's payoff over *strategic messaging* alone, where the receiver keeps his correctly specified model.¹⁴

Definition 2. The sender *fully manipulates* the receiver if he chooses $a = 1$ with probability 1, *partially manipulates* the receiver if he chooses $a = 1$ with probability strictly less than 1 but strictly greater than the probability attainable in any equilibrium with strategic messaging alone, and *cannot manipulate* otherwise.

¹³For conditions (iii) and (iv), “on the equilibrium path” means that $\hat{\sigma} = \hat{\sigma}^*, \sigma = \sigma^*(\hat{\sigma}^*)$, $m \in \text{supp}(\sigma^*(\cdot|\omega, \hat{\sigma}^*)) \cup \text{supp}(\hat{\sigma}^*(\cdot|\omega))$ for some $\omega \in \Omega$. Off the equilibrium path (i.e., following a deviation $\hat{\sigma}' \neq \hat{\sigma}^*$ or m' outside this union for every $\omega \in \Omega$), the receiver's model-switching rule and posterior beliefs are left unrestricted. Conditions (i) and (ii) require optimality given some specified off-path objects, consistent with weak PBE.

¹⁴While we focus on the gain from adding narratives to strategic messaging, Supplemental Appendix B.4.5 examines the gain from optimized messaging relative to narrative persuasion alone.

2.2 Discussion of Assumptions

Interpretation of the proposed model Although the proposed model and the initial model may appear asymmetric—one summarized by a single $\hat{\sigma}$ and the other a contingent plan $(\hat{\sigma}, \sigma(\cdot))$ —they are symmetric in the following sense: the proposed model claims that the sender proposes and then follows $\hat{\sigma}$, as if she had commitment power, so that $\hat{\sigma}$ is a sufficient description of her strategy.¹⁵ Note that $\hat{\sigma}$ is a behavioral description rather than a strategically derived equilibrium strategy. Many different primitives could rationalize the same proposed strategy, such as commitment types, intrinsic preferences, or institutional constraints. Since the receiver’s inference depends only on the implied messaging strategy, this reduced-form approach allows us to isolate the persuasion mechanism while remaining agnostic about its microfoundation.

Why does the receiver switch at all? If the receiver knew for certain his initial model was correct, he would realize that allowing model switching could only lower his equilibrium payoff.¹⁶ For a non-dogmatic receiver facing potentially diverse sender types, however, model switching can be a natural response—indeed, if the sender were the type she claims to be, switching to the proposed model would allow the receiver to act on correct beliefs and weakly improve his expected payoff. He may therefore consider alternatives without the knowledge or capacity to specify the full set of possible sender types, as formalized in the literature on unawareness and limited consideration (Dekel et al., 1998; Gennaioli and Shleifer, 2010). We accordingly model him as focusing on his initial model and the salient alternative proposed by the sender. This formulation provides a disciplined way to study how a strategic sender can exploit reasonable openness to alternative models, even when the receiver’s initial model is correct.

Switching versus averaging Anecdotal and experimental evidence suggests that people often use one model at a time and switch when evidence sufficiently favors an alternative, rather than updating continuous probabilistic beliefs over them with Bayes’ rule (Barron and Fries, 2025; Aina and Schneider, 2025). This may reflect cognitive constraints on reasoning across structurally distinct models. Our switching framework delivers sharp results, with the extent of manipulation cleanly characterized by the skepticism parameter α . A more Bayesian approach would instead have the receiver assign a positive prior weight to the proposed model before hearing any message, which may be less plausible for narratives

¹⁵In the initial model’s format, the proposed model is $(\hat{\sigma}, \sigma'(\cdot))$ where $\sigma'(\hat{\sigma}) = \hat{\sigma}$ for all $\hat{\sigma}$.

¹⁶This claim is nontrivial for general information technologies; see Proposition B.2 in Supplemental Appendix B.3.4.

the receiver had not previously considered. We develop the comparison further in [Section 6](#).

(In)correctly specified initial model We assume the receiver’s initial model is correctly specified to minimize departures from standard communication models and stack the environment against the sender.¹⁷ [Section 6](#) discusses how initial misspecification creates further opportunities for manipulation.

3 The Cheap Talk Benchmark

We begin with the cheap talk benchmark, where messages are unconstrained. Strikingly, although cheap talk is powerless in a standard setting, narratives make full manipulation possible under economically intuitive conditions.

3.1 Without Narratives

Consider what the sender can achieve through strategic messaging alone, with or without commitment power. Given a messaging strategy σ used by the receiver for inference, define the sets of *winning* messages as $M_\sigma^+ := \{m \in \text{supp}(\sigma) : \mathbb{E}_{\omega \sim \pi_\sigma(\cdot|m)}[\omega] \geq \omega_0\}$, and *losing* messages as $M_\sigma^- := \text{supp}(\sigma) \setminus M_\sigma^+$. Winning messages induce the risky action, while losing messages induce the safe action. A sender with commitment solves the Bayesian persuasion (BP) problem to maximize ex ante winning probability:

$$\max_{\sigma \in \Sigma} \int_{\Omega} \sigma(M_\sigma^+|\omega) dF(\omega). \quad (8)$$

The solution has the familiar cutoff structure ([Kamenica and Gentzkow, 2011](#)). Let

$$\omega^* := \max\{\tilde{\omega} : \mathbb{E}_{\omega \sim F}[\omega|\omega \geq \tilde{\omega}] \leq \omega_0\}. \quad (9)$$

An optimal policy sends a favorable message H for $\omega > \omega^*$, an unfavorable one L for $\omega < \omega^*$, and may mix at $\omega = \omega^*$ so that the receiver is indifferent after H and strictly prefers the safe action after L . Let $p^* \in (0, 1)$ denote the resulting winning probability. When F is atomless, the cutoff satisfies $\mathbb{E}_{\omega \sim F}[\omega|\omega \geq \omega^*] = \omega_0$ and $p^* = F(\omega \geq \omega^*)$.¹⁸

Lemma 1. *In any solution to the BP problem ([Eq. \(8\)](#)), almost all types $\omega > \omega^*$ win, almost all types $\omega < \omega^*$ lose, and type ω^* may mix, yielding winning probability p^* .*

¹⁷One interpretation is that the receiver learns the sender’s strategy through experience with similar senders or infers it from their incentives.

¹⁸If F has an atom at cutoff ω^* , the sender sends H with probability q and L otherwise, where $q = \frac{\int_{(\omega^*, \sup \Omega]} \omega dF(\omega) - \omega_0 F(\omega > \omega^*)}{(\omega_0 - \omega^*) F(\{\omega^*\})}$ when $\omega^* < \omega_0$ and $q = 1$ otherwise. In this case, the resulting winning probability is $p^* = F(\omega > \omega^*) + F(\{\omega^*\}) \cdot q$.

Without commitment, the familiar unraveling logic of Crawford and Sobel (1982) applies: any winning message would be flooded by losing types, causing communication to collapse into babbling in equilibrium, so the sender's maximum winning probability is $\tilde{p}^* := 0$. Adding narratives could potentially bridge the gap between $\tilde{p}^* = 0$ and $p^* \in (0, 1)$, but as we will now see, this is a vast understatement.

3.2 With Narratives

We now allow the sender to also propose a narrative $\hat{\sigma} \in \bar{\Sigma}$ alongside controlling the messaging strategy. Her ability to manipulate the receiver turns out to hinge on a simple demand-budget comparison. Define the *demand for narratives* given messaging strategy σ as the probability of sending a losing message under σ ,

$$D(\sigma) := \int_{\Omega} \sigma(M_{\sigma}^{-}|\omega) dF(\omega), \quad (10)$$

which is bounded below by $D := 1 - p^*$ by Lemma 1. We refer to D as the (minimum) narrative demand because it is the minimum probability mass of types that lose through messaging and therefore need to win through model switching. Analogously, define the *budget for narratives* given proposed model $\hat{\sigma}$ as its implied winning probability,

$$B(\hat{\sigma}) := \int_{\Omega} \hat{\sigma}(M_{\hat{\sigma}}^{+}|\omega) dF(\omega), \quad (11)$$

which is bounded above by $B := p^*$ by Lemma 1. We refer to B as the (maximum) narrative budget because it is the largest probability a proposed model can assign to winning messages. Theorem 1 shows a sharp dichotomy: the sender either fully manipulates the receiver or babbles, depending on whether the budget covers the demand.

Theorem 1. *A cheap-talk sender fully manipulates in the sender-optimal equilibrium if and only if $\alpha D \leq B$, or equivalently $p^* \geq \alpha/(1 + \alpha)$, and cannot manipulate otherwise.*

Full manipulation requires the narrative budget B to cover demand D at price α : to convince the receiver to switch models on any message m , the sender must send m at least α times as often under $\hat{\sigma}$ as under σ , so each unit of demand costs α units of budget. The equivalent expression $p^* \geq \alpha/(1 + \alpha)$ yields intuitive comparative statics. Greater skepticism (higher α) raises the price and tightens the budget constraint, whereas a more favorable prior (higher p^*) relaxes it by raising the budget and lowering the demand. When $\alpha = 1$, full manipulation requires that BP alone let the sender win at least half the time. For any fixed prior, full manipulation becomes feasible as $\alpha \rightarrow 0$ and infeasible as $\alpha \rightarrow \infty$.

The sender can implement full manipulation through a confidence trick, illustrated in

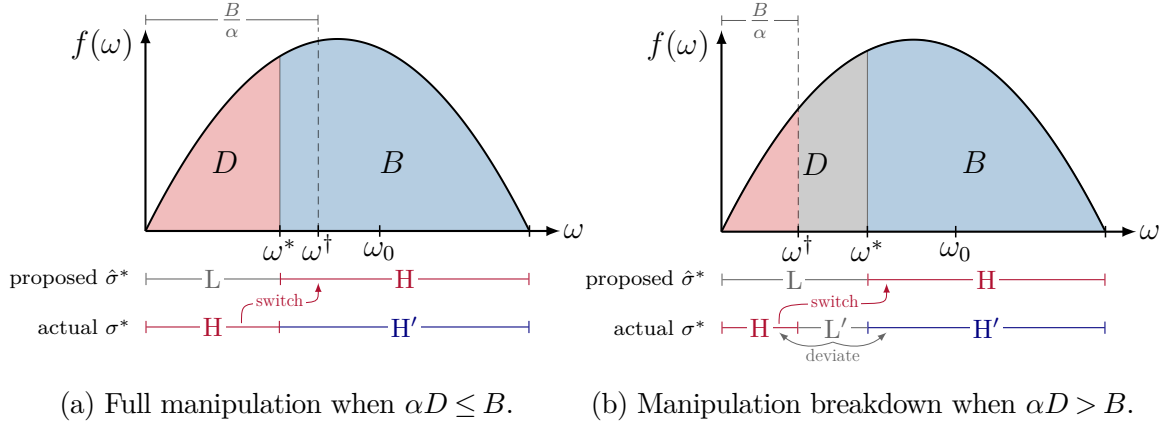


Figure 1: Manipulation with cheap talk and narratives.

Figure 1a.¹⁹ She proposes the narrative that she is a “Bayesian persuader,” sending H for $\omega \geq \omega^*$ and L for $\omega < \omega^*$, effectively claiming commitment power she does not have. In actuality she does the opposite: she sends H for $\omega < \omega^*$ and H' for $\omega \geq \omega^*$. After H , the receiver switches models because H is sent with probability D under the true strategy and with probability B under the proposed model, so $\lambda(H) = B/D \geq \alpha$. He then forms a posterior that puts all weight on $\omega \geq \omega^*$ and incorrectly expects $\mathbb{E}[\omega | \omega \geq \omega^*] = \omega_0$, which leads to the risky action even though it is strictly worse because the true state satisfies $\omega < \omega^*$. By sending H' , higher types strategically contradict their own narrative as H' is never sent under $\hat{\sigma}$, so $\lambda(H') = 0$ and the receiver retains his original model. He then correctly expects $\mathbb{E}[\omega | \omega \geq \omega^*] = \omega_0$ and takes the risky action. Thus every type induces the risky action, and since no type has a profitable deviation, this is indeed an equilibrium.

Note that full manipulation per se does not require the sender to pose as a Bayesian persuader. She could instead claim to be honest, for example, recommending the risky action only when $\omega \geq \omega_0$. This yields a smaller narrative budget $B_H := F(\omega \geq \omega_0) \leq B$, and the same construction achieves full manipulation whenever $B_H \geq \alpha D$. Thus, the Bayesian-persuasion narrative is optimal, but not essential.

However, our construction uses the narrative budget as efficiently as possible. It yields $\lambda(H) = B/D$ by allocating the entire narrative budget to H , subject to the condition that the receiver weakly prefers the risky action following any on-path message. When $B < \alpha D$, manipulation breaks down entirely (Figure 1b). The BP narrative fails because the receiver no longer believes it after H . To restore switching, the sender could instead send H only for $\omega \leq \omega^\dagger$ where $\omega^\dagger < \omega^*$ is chosen so that $\lambda(H) = \alpha$. But this leaves losing

¹⁹Figure 1 assumes a continuous state distribution; with discrete states, the sender may mix at cutoffs.

types in $(\omega^\dagger, \omega^*)$ who can profitably deviate to a winning message, causing the strategy to unravel. No other allocation resolves this problem: the aggregate demand–budget bound shows that the budget is exhausted before all types can be covered, leaving a residual interval of deviators. Babbling is thus the unique outcome.²⁰

Theorem 1 reveals a novel complementarity between strategic messaging and narrative persuasion. Strategic messaging alone yields 0, while narrative persuasion itself wins at most an extra fraction p^*/α of states, both of which could be far below the BP payoff p^* . Yet when used jointly, the sender leaps all the way to payoff 1, fully manipulating a receiver who began with a correctly specified model. This is because narrative persuasion not only enables losing types to win, but also eliminates the credibility hurdle that prevents informative communication in the first place: since every type wins, no type has a profitable deviation, so the usual unraveling logic disappears. This in turn frees strategic messaging to operate at full force, replicating the BP payoff and then surpassing it.

3.3 Welfare Implications

The characterization generates two stark welfare implications. First, full manipulation can generate *arbitrarily large* welfare loss for the receiver.

Corollary 1. *Fixing any α , ω_0 , and $U > 0$, there exists a state distribution F under which full manipulation is feasible and the receiver obtains $\mathbb{E}_{\omega \sim F}[\omega] < -U$.*

The following example illustrates and proves the result.

Example 1. (Picking up pennies in front of a steamroller²¹) Suppose $\alpha = 1$, $\omega_0 = 0$, and the risky action yields either $\varepsilon > 0$ or $-b < 0$ with equal probability. As $\varepsilon \rightarrow 0$ and $-b \rightarrow -\infty$, the risky action offers a vanishing upside against an arbitrarily large downside. Nevertheless, since $p^* > 1/2 = \alpha/(1 + \alpha)$, full manipulation is feasible for any b by **Theorem 1**. The sender can therefore always convince the receiver to take the risky action, even as its expected payoff approaches $-\infty$. The same logic extends to any α : if the risky action yields $\omega_0 + \varepsilon$ with probability $\alpha/(1 + \alpha)$ and $-b$ otherwise, full manipulation remains feasible as $-b \rightarrow -\infty$.

Second, while the sender always benefits from a more favorable state distribution, the receiver’s welfare is non-monotone. To see this, consider shifting the state distribution F

²⁰A babbling equilibrium exists as follows: the sender proposes $\hat{\sigma}^*$ that pools all types on a single message, say H , and plays $\sigma^*(H|\omega, \hat{\sigma}) = 1$ for all ω and all $\hat{\sigma}$ on and off path. On path, both models yield the prior, so the receiver takes the safe action whether or not he switches. Off-path beliefs induce the safe action after every deviation, leaving the sender no profitable deviation.

²¹**Taleb (2007)** uses this term to describe a bet with negligible upside and catastrophic downside.

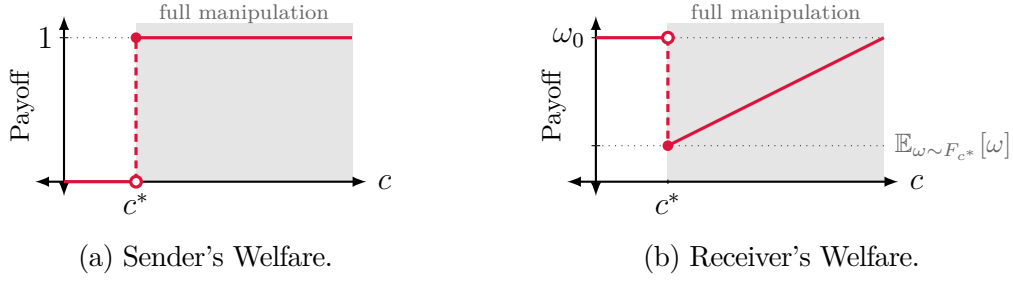


Figure 2: Equilibrium welfare comparison as c increases.

by some constant $c \in \mathbb{R}$. Let F_c denote the distribution with density $f_c(\omega + c) := f(\omega)$ for all $\omega \in \Omega$ while keeping ω_0 fixed and maintaining the assumptions $\mathbb{E}_{\omega \sim F_c}[\omega] < \omega_0$ and $F_c(\omega \geq \omega_0) > 0$. An upward shift allows more types to pool and win, weakly increasing the BP winning probability p_c^* and relaxing the budget-demand condition.

Corollary 2. *There exists a threshold c^* such that in the sender-optimal equilibrium,*

- (i) *the sender's payoff is 0 for $c < c^*$ and 1 for $c \geq c^*$;*
- (ii) *the receiver's payoff is ω_0 for $c < c^*$, drops discontinuously to $\mathbb{E}_{\omega \sim F_{c^*}}[\omega] < \omega_0$ at c^* , and increases linearly in c thereafter but remains under ω_0 .*

Figure 2 illustrates the result. A better distribution facilitates full manipulation and weakly benefits the sender, but can harm the receiver. For $c < c^*$, manipulation is infeasible and the receiver obtains the safe payoff ω_0 . At c^* , full manipulation becomes feasible, and his payoff drops to the unconditional state mean, strictly below ω_0 and potentially arbitrarily low (Corollary 1). Beyond c^* , his payoff rises linearly in c but remains below ω_0 by assumption.

4 General Information Technology

We now generalize the analysis beyond cheap talk to a broad class of information technologies and study how information hardness affects the extent of manipulation. As before, we first examine the case without narratives and then the equilibrium with narratives. We focus on a continuously distributed state throughout this section.²²

Following Hart et al. (2017) and Rappoport (2025), we take the feasible message set to be a subset of the type space, so that sending message ω' in state ω is interpreted as type ω mimicking ω' . To vary information hardness continuously, we focus on *monotone interval technologies*, in which type ω can claim any state $\omega' \in M(\omega) = [L(\omega), U(\omega)]$,

²²The finite Ω case is more notation-heavy and is handled in Supplemental Appendix B.3.3.

where $L, U : \Omega \rightarrow \Omega$ are weakly increasing with $L(\omega) \leq \omega \leq U(\omega)$. This captures the natural idea that higher states have access to higher claims, and that the sender can always report truthfully. The wider the interval $[L(\omega), U(\omega)]$, the more types a sender can mimic and the softer her information is. For tractability, we assume L and U are continuously differentiable except at finitely many points, with L left- and U right-continuous.²³ Denote the set of all such technologies as $\mathbb{M}$.

Example 2. Examples of monotone interval technologies in $\mathbb{M}$ include:

- (i) *Cheap talk*: $L(\omega) = \inf \Omega$ and $U(\omega) = \sup \Omega$, so $M(\omega) = \Omega$.
- (ii) *Mandatory disclosure*: $L(\omega) = U(\omega) = \omega$, so $M(\omega) = \{\omega\}$.
- (iii) *Understating*: $L(\omega) \leq \omega$ and $U(\omega) = \omega$, so $M(\omega) = [L(\omega), \omega]$. The sender can underreport the state down to $L(\omega)$ but not overreport.
- (iv) *Overstating*: $L(\omega) = \omega$ and $U(\omega) \geq \omega$, so $M(\omega) = [\omega, U(\omega)]$. The sender can overreport the state up to $U(\omega)$ but not underreport.
- (v) *Coarse evidence*: let $\{\Omega_1, \dots, \Omega_N\}$ cover Ω with $N \geq 2$ non-degenerate closed intervals having disjoint interiors, and set $M(\omega) = \bigcup_{n:\omega \in \Omega_n} \Omega_n$. The sender can mimic types sharing the same evidence; at a shared boundary, she can present either adjacent piece of evidence.

Remark 1. The disclosure literature often allows the sender to remain silent by sending $\emptyset$ (e.g., [Grossman, 1981](#); [Milgrom, 1981](#)). Since silence is available to every type, it acts as a universal pooling device that can substantially change the sender's persuasiveness in our setting. We analyze this case separately in Supplemental Appendix B.2.

4.1 Without Narratives

As before, we first characterize what the sender can achieve through strategic messaging alone, with or without commitment. The monotone interval structure allows us to obtain a clean characterization due to two key properties formalized in [Lemma 2](#).

Lemma 2. Let $\Omega_{\mathcal{M}}(m)$ denote the set of types that can send m under $\mathcal{M}$.

- (i) For any $m \in \Omega$, $\Omega_{\mathcal{M}}(m)$ is a closed interval.
- (ii) For any $m \in \Omega$, $\omega_1, \omega_2 \in \Omega_{\mathcal{M}}(m)$ with $\omega_1 < \omega_2$, and any $m' \in \Omega$ that can be sent by some type $\omega \in [\omega_1, \omega_2]$, at least one of ω_1 and ω_2 can also send m' .

²³The one-sided continuity conditions require $L(\omega) = \lim_{\omega' \uparrow \omega} L(\omega')$ and $U(\omega) = \lim_{\omega' \downarrow \omega} U(\omega')$, which together with monotonicity ensure that the correspondence $M(\omega) = [L(\omega), U(\omega)]$ has a closed graph.

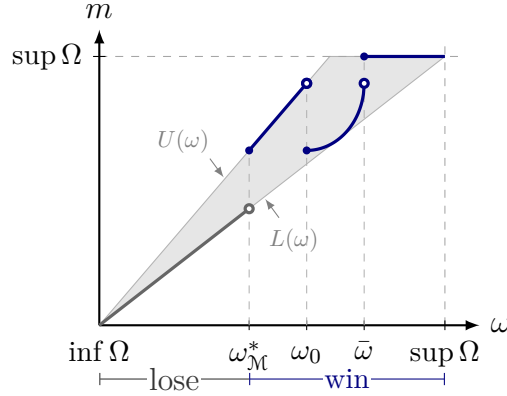


Figure 3: Solution to the constrained BP problem under an overstating technology.²⁴ The gray band depicts the feasible message set $[L(\omega), U(\omega)]$ and the solid curve shows the optimal messaging strategy (dark gray for losing messages and blue for winning messages).

With commitment, the sender solves a Bayesian persuasion problem subject to the messaging constraints (Eq. (8)). **Proposition 1** shows that it retains a cutoff structure.

Proposition 1. *Fix $\mathcal{M} \in \mathbb{M}$. Any solution to the constrained BP problem has a cutoff structure: there exists $\omega_{\mathcal{M}}^* \in \Omega$ with $\omega_{\mathcal{M}}^* \leq \omega_0$ such that almost all types $\omega \geq \omega_{\mathcal{M}}^*$ win and almost all types $\omega < \omega_{\mathcal{M}}^*$ lose, yielding winning probability $p_{\mathcal{M}}^* := F(\omega \geq \omega_{\mathcal{M}}^*)$.*

All solutions share the same cutoff $\omega_{\mathcal{M}}^*$ and winning probability $p_{\mathcal{M}}^*$; **Appendix A.3.2** provides a formal construction that can be used to compute them. **Fig. 3** illustrates one solution under an overstating information technology with $U(\omega) = \min\{\delta\omega, \sup \Omega\}$ for $\delta > 1$. Low types in $[\omega_{\mathcal{M}}^*, \omega_0)$ send their highest feasible message and pool with high types in $[\omega_0, \bar{\omega})$, whose strategy is calibrated so that each pooled message induces a posterior mean equal to ω_0 , making the receiver indifferent. High types are exhausted from ω_0 upward because types closer to ω_0 send lower messages and are thus more accessible to low types, allowing as many low types as possible to win. Types below $\omega_{\mathcal{M}}^*$ report truthfully and lose, while types above $\bar{\omega}$ send their highest feasible messages and win. The cutoff $\omega_{\mathcal{M}}^*$ is the lowest type for which this construction remains feasible: a lower cutoff requires more low types to win by pooling with high types, either exhausting all high types or pushing the construction further down until it hits the lower bound L .

We now turn to the equilibrium without commitment, where the sender must also ensure that no type wants to deviate. The key question is then whether the lowest winning type can credibly separate from lower types by sending a message they cannot

²⁴The figure uses $\Omega = [0, 1]$, $\omega_0 = 0.6$, $L(\omega) = \omega$, $U(\omega) = \min\{1.5\omega, 1\}$, and a uniform prior. For a full derivation of the depicted solution, see Lemma B.2 in Supplemental Appendix B.3.1.

access. Under cheap talk this is impossible, so only babbling can be sustained. For general technologies, however, feasibility constraints can sustain credible separation. We call a type *credible* if it can send a message that no lower type can send.

Definition 3. Type ω is *credible* under $\mathcal{M}$ if $U(\omega) > U(\omega')$ for any $\omega' < \omega$.

Let $\tilde{\omega}_{\mathcal{M}}^*$ be the infimum of credible types at or above $\omega_{\mathcal{M}}^*$, and set $\tilde{p}_{\mathcal{M}}^* := F(\omega \geq \tilde{\omega}_{\mathcal{M}}^*)$. If no such type exists, leave $\tilde{\omega}_{\mathcal{M}}^*$ undefined and set $\tilde{p}_{\mathcal{M}}^* = 0$. **Proposition 2** establishes that $\tilde{p}_{\mathcal{M}}^*$ is the sender's optimal equilibrium payoff without commitment.

Proposition 2. Fix $\mathcal{M} \in \mathbb{M}$. In the pure messaging game, the sender-optimal equilibrium winning probability is $\tilde{p}_{\mathcal{M}}^* \leq p_{\mathcal{M}}^*$. If $\tilde{\omega}_{\mathcal{M}}^*$ exists, it is attained by an equilibrium in which types $\omega > \tilde{\omega}_{\mathcal{M}}^*$ win and types $\omega < \tilde{\omega}_{\mathcal{M}}^*$ lose. If $\tilde{\omega}_{\mathcal{M}}^*$ does not exist, all types lose in every sender-optimal equilibrium.

Note that the cutoff $\tilde{\omega}_{\mathcal{M}}^*$ may not be credible; when it is not, it loses in every sender-optimal equilibrium. Since F is continuous, this does not affect the winning probability. The absence of commitment power can only weakly reduce the sender's payoff in the pure messaging game, as reflected in $\tilde{p}_{\mathcal{M}}^* \leq p_{\mathcal{M}}^*$. In the overstating example seen in **Fig. 3**, $\omega_{\mathcal{M}}^*$ is itself credible, so $\tilde{\omega}_{\mathcal{M}}^* = \omega_{\mathcal{M}}^*$ and the loss of commitment power does not reduce the sender's payoff.

To the best of our knowledge, **Propositions 1** and **2** provide the first explicit characterization of the BP solution and the sender-optimal equilibrium without narratives for the full class of monotone interval technologies.²⁵

4.2 With Narratives

We now allow the sender to jointly control the narrative and her messaging strategy. As in cheap talk, full manipulation hinges on whether the narrative budget can fully cover the demand. **Theorem 2** extends this condition to monotone interval technologies.

Theorem 2 (Full Manipulation). Fix any $\mathcal{M} \in \mathbb{M}$. The sender fully manipulates in sender-optimal equilibrium if and only if $B \geq \alpha D_{\mathcal{M}}$, or equivalently $p^*/\alpha \geq 1 - p_{\mathcal{M}}^*$.

Since strategic messaging with commitment wins at most $p_{\mathcal{M}}^*$, narrative demand is now $D_{\mathcal{M}} := 1 - p_{\mathcal{M}}^*$. The budget remains at $B = p^*$ since the proposed model is

²⁵Relatedly, **Antić and Chakraborty (2026)** characterize optimal persuasion under commitment as a matching problem, allowing general evidence constraints and state-dependent sender preferences in finite state-message spaces. They also identify cases in which commitment has no value. Our monotone interval structure and state-independent sender preferences yield explicit cutoff characterizations with and without commitment for continuous states. We provide finite-state results in Supplemental Appendix B.3.3.

unconstrained. Complementarity again plays a key role here: narrative persuasion removes the credibility hurdle that otherwise caps the sender’s messaging payoff at $\tilde{p}_{\mathcal{M}}^*$, freeing strategic messaging to contribute its full commitment value $p_{\mathcal{M}}^*$, thereby achieving full manipulation whenever $p^*/\alpha + p_{\mathcal{M}}^* \geq 1$. Notably, this condition can hold under mandatory disclosure ($M(\omega) = \{\omega\}$), where the sender must reveal evidence that uniquely identifies the state, provided $p^*/\alpha \geq F(\omega < \omega_0)$.

One strategy achieving full manipulation generalizes the confidence trick from [Section 3](#). In this construction, the sender’s actual messaging strategy σ^* replicates the constrained BP solution from [Proposition 1](#): high types $\omega \geq \omega_{\mathcal{M}}^*$ send winning messages and low types $\omega < \omega_{\mathcal{M}}^*$ send losing messages. In the proposed model $\hat{\sigma}^*$, the sender *flips the interpretation* of losing messages by claiming that they are sent by types $\omega \geq \omega^*$, where ω^* is the unconstrained BP cutoff. When $B \geq \alpha D_{\mathcal{M}}$, these messages induce switching and are interpreted as good news. Winning messages are never sent under $\hat{\sigma}^*$, so the receiver retains his initial model after observing them and correctly infers that only high types send them. This construction is efficient: the actual strategy minimizes narrative demand at $D_{\mathcal{M}}$, while the proposed model allocates the entire budget B to messages that lose under the initial model. Full manipulation is therefore feasible if and only if the budget B suffices to cover the demand $D_{\mathcal{M}}$ at price α .

Remark 2. Although we focus on monotone interval technologies, [Theorem 2](#) extends to any technology that admits an optimal constrained BP solution.²⁶ The monotone interval structure is used to characterize $p_{\mathcal{M}}^*$ and $\tilde{p}_{\mathcal{M}}^*$ explicitly via [Propositions 1](#) and [2](#); it becomes more essential for the partial manipulation characterization that follows.

When full manipulation is infeasible, message constraints may sustain partial manipulation by blocking low types from imitating winning types, allowing the sender to make the most of both narrative persuasion and strategic messaging. Again, the relevant question is whether a credible cutoff exists that stabilizes the losing region.

For a clean characterization of partial manipulation, we restrict to *regular* monotone interval technologies. We call $\mathcal{M} = (L, U) \in \mathbb{M}$ regular if L and U are continuous on Ω and strictly increasing wherever they take values in its interior, and denote this subclass by $\mathbb{M}^* \subset \mathbb{M}$.²⁷ Under regularity, credibility is simply determined by whether a type can

²⁶For instance, under canonical set-valued disclosure (e.g., [Grossman, 1981](#)), where the sender in state ω can disclose any set $S \ni \omega$, the BP solution pools all states above ω^* . So $p_{\mathcal{M}}^* = p^*$ and $D_{\mathcal{M}} = D$, and full manipulation is feasible if and only if $B \geq \alpha D$, the same condition as under cheap talk.

²⁷Regularity accommodates cheap talk and mandatory disclosure in [Example 2](#)(i)–(ii), and the understating and overstating technologies in (iii)–(iv) when their bounds satisfy these conditions. Without

mimic every higher type. Define the infimum of such types as

$$c_{\mathcal{M}} := \inf\{\omega \in \Omega : U(\omega) = \sup \Omega\}, \quad (12)$$

with $c_{\mathcal{M}} = +\infty$ if the set is empty; each $\omega \leq c_{\mathcal{M}}$ is credible, while each $\omega > c_{\mathcal{M}}$ is not.

Theorem 3 shows that harder information can generate a wide range of manipulation outcomes. When $B < \alpha D_{\mathcal{M}}$, the narrative budget cannot cover the full demand, so some types must lose. A natural strategy is to economize on the limited budget by giving up on reinterpreting the lowest types' messages and running the confidence trick only on intermediate types. Define $\omega_{\mathcal{M}}^{**}$ as the unique type at which the budget exactly covers the residual demand $D_{\mathcal{M}} - F(\omega < \omega_{\mathcal{M}}^{**})$ at price α , assuming losing types concentrate below $\omega_{\mathcal{M}}^{**}$. Its position relative to the credibility boundary $c_{\mathcal{M}}$ determines which instruments the sender can use in equilibrium, leading to two distinct regimes.

Theorem 3 (Partial and No Manipulation). *Fix $\mathcal{M} \in \mathbb{M}^*$ with $B < \alpha D_{\mathcal{M}}$, and let $\omega_{\mathcal{M}}^{**}$ be defined by*

$$B = \alpha(D_{\mathcal{M}} - F(\omega < \omega_{\mathcal{M}}^{**})). \quad (13)$$

- (i) *If $\omega_{\mathcal{M}}^{**} < c_{\mathcal{M}}$, a sender-optimal equilibrium exists where types $\omega \geq \omega_{\mathcal{M}}^{**}$ win and $\omega < \omega_{\mathcal{M}}^{**}$ lose. The sender partially manipulates with winning probability $p^*/\alpha + p_{\mathcal{M}}^*$.*
- (ii) *If $\omega_{\mathcal{M}}^{**} \geq c_{\mathcal{M}}$, the sender partially manipulates with winning probability p^*/α in a sender-optimal equilibrium if and only if $B \geq \alpha F(\Omega_{\mathcal{M}}(m^*))$ for some $m^* \in \Omega$.*
- (iii) *Otherwise, all types lose in every equilibrium and the sender cannot manipulate.*

First, when $\omega_{\mathcal{M}}^{**} < c_{\mathcal{M}}$, the sender wins by combining strategic messaging and narrative persuasion. **Figure 4a** illustrates an equilibrium using the overstating technology with $M(\omega) = [\omega, \delta\omega] \cap \Omega$ and $\delta > 1$. Types above $\omega_{\mathcal{M}}^*$ follow the constrained BP strategy and win under the receiver's initial model. Intermediate types in $[\omega_{\mathcal{M}}^{**}, \omega_{\mathcal{M}}^*)$ win via narrative persuasion: the sender claims that high types would send the same messages these intermediate types send, so the receiver reinterprets them as good news. Types below $\omega_{\mathcal{M}}^{**}$ report truthfully and lose. Since $\omega_{\mathcal{M}}^{**} < c_{\mathcal{M}}$, the cutoff is credible, so the construction can place every winning message beyond the reach of lower types. The complementarity between messaging and narratives is especially stark in this example: pure messaging yields zero by **Proposition 2**, since no type at or above $\omega_{\mathcal{M}}^*$ is credible, whereas their joint use delivers $F(\omega \geq \omega_{\mathcal{M}}^{**}) = p_{\mathcal{M}}^* + p^*/\alpha$.

regularity, selecting messages for narrative persuasion can involve a knapsack-type problem; Supplemental Appendix B.3.2 solves the coarse-evidence case in (v).

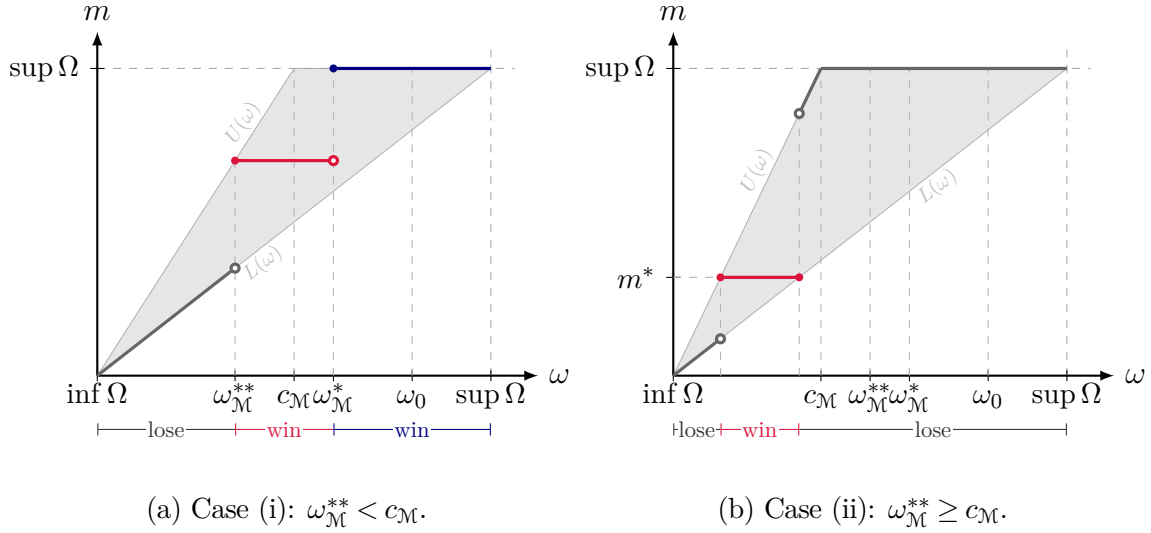


Figure 4: Partial manipulation with regular monotone interval technologies. The gray band depicts $[L(\omega), U(\omega)] \cap \Omega$ and the colored curves show σ^* (dark gray for losing messages, red for those that win via a model switch, and blue for those that win under the initial model).

Second, when $\omega_M^{**} \geq c_M$, the sender can only win through narrative persuasion. As illustrated in Fig. 4b using the overstating technology with a higher δ , the previous strategy fails: if $\omega_M^{**} > c_M$, lower types can imitate higher types above ω_M^{**} ; if $\omega_M^{**} = c_M$, only the highest message $\max \Omega$ is inaccessible to types below the cutoff, leaving no room to separate messages that win under the initial model from those that win through switching. Instead, the sender looks for a message m^* whose feasible pool $\Omega_M(m^*)$ is small enough for the narrative budget to cover. If such a message exists, regularity allows her to expand the winning message interval around m^* until the budget is exhausted. Types in the resulting feasible pools send the selected messages, while the proposed model claims that types above ω^* would send those same messages. The receiver consequently switches models and takes the risky action after each selected message, achieving winning probability p^*/α . If no such message exists, the budget cannot support any winning message through switching and all types lose.

We conclude this section with two observations on the value of narratives. Narratives weakly improve on the sender's optimal equilibrium payoff and strictly do so whenever she fully or partially manipulates. Moreover, whenever there is complementarity between messaging and narratives, the sender strictly surpasses the BP benchmark. Under full manipulation the sender obtains payoff 1, and under partial manipulation with $\omega_M^{**} < c_M$ she obtains $p_M^* + p^*/\alpha$, both strictly higher than p_M^* and $\tilde{p}_M^*$. When $\omega_M^{**} \geq c_M$, complementarity breaks down and partial manipulation yields p^*/α , which strictly exceeds the zero payoff without narratives but need not exceed p_M^* .

Corollary 3. *Under full manipulation or partial manipulation with $\omega_{\mathcal{M}}^{**} < c_{\mathcal{M}}$, the sender's payoff strictly exceeds the BP payoff $p_{\mathcal{M}}^*$.*

5 Information Technology and Persuasiveness

We now use the equilibrium characterization to study how information technology shapes the sender's payoff. We first examine the effects of greater flexibility and then characterize the technologies that maximize the sender's payoff across environments.

5.1 Flexibility and Credibility

Our analysis centers on a tension between two properties: *flexibility*, which determines how broadly types can pool on the same message, and *credibility*, which determines how effectively higher types can distinguish themselves from lower types. This tension also arises in disclosure games without narratives.²⁸ We compare the effects of flexibility with and without narratives to identify how narrative persuasion reshapes the tradeoff. Because downward and upward expansions of the message set have different implications for credibility, we distinguish between the two.

Definition 4. Given two information technologies $\mathcal{M} = (L, U)$ and $\mathcal{M}' = (L', U')$ in $\mathbb{M}$, $\mathcal{M}'$ is *more flexible* than $\mathcal{M}$ if $M(\omega) \subseteq M'(\omega)$ for all $\omega \in \Omega$. In particular, $\mathcal{M}'$ is *more downward flexible* than $\mathcal{M}$ if $U'(\omega) = U(\omega)$ and $L'(\omega) \leq L(\omega)$ for all $\omega \in \Omega$, and *more upward flexible* if $L'(\omega) = L(\omega)$ and $U'(\omega) \geq U(\omega)$ for all $\omega \in \Omega$.

A more flexible technology can benefit the sender by raising the BP payoff $p_{\mathcal{M}}^*$ and thereby lowering the narrative demand $D_{\mathcal{M}} = 1 - p_{\mathcal{M}}^*$, but may also erode credibility by making more types non-credible. However, the potential credibility cost is irrelevant for full manipulation as it depends only on the demand-budget comparison ([Theorem 2](#)).

Corollary 4. *Fix $\mathcal{M}, \mathcal{M}' \in \mathbb{M}$, with $\mathcal{M}'$ more flexible than $\mathcal{M}$. If full manipulation is feasible under $\mathcal{M}$, it is also feasible under $\mathcal{M}'$.*

Greater flexibility cannot undo full manipulation once it is feasible, but without narratives it can reduce the sender's payoff to zero by eliminating credible separation ([Fig. 5a](#)). These effects can dramatically reverse the sender's ranking of information technologies.

²⁸For instance, [Lizzeri et al. \(2026\)](#) study this tension in a sender–receiver game with partially aligned preferences under either cheap talk or perfect disclosure. Flexibility benefits the sender when preferences are sufficiently aligned and harms her otherwise. In our setting, flexibility can benefit the sender despite her state-independent preference for the risky action.

Without narratives, cheap talk yields zero and is strictly worse than mandatory disclosure. With narratives, cheap talk permits full manipulation over a wider range of parameters and strictly outperforms mandatory disclosure whenever only the former permits full manipulation. This provides a theoretical rationale for why narratives and soft or unverifiable information can be especially persuasive together.

When full manipulation is infeasible, the direction of flexibility matters. We again restrict attention to regular technologies so that [Theorem 3](#) applies. Without narratives, downward flexibility lets higher types reach lower messages and pool with more lower types without affecting their ability to separate from them, so it weakly benefits the sender. Upward flexibility also expands pooling opportunities but allows lower types to imitate higher types. It harms the sender precisely when this loss of credible separation reduces her payoff to zero. With narratives, flexibility affects the credibility of both strategic messaging and narrative persuasion. The following corollary characterizes these effects. For brevity, say that $\mathcal{M}$ satisfies the *feasible-pool condition* if $B \geq \alpha F(\Omega_{\mathcal{M}}(m))$ for some $m \in \Omega$. As shown in [Theorem 3\(ii\)](#), this condition is necessary for partial manipulation through narrative persuasion alone when $\omega_{\mathcal{M}}^{**} \geq c_{\mathcal{M}}$.

Corollary 5. *Suppose the sender's technology changes from $\mathcal{M}$ to $\mathcal{M}'$, both in $\mathbb{M}^*$.*

- (i) *Suppose $\mathcal{M}'$ is more downward flexible than $\mathcal{M}$.*
 - (a) *Without narratives, the sender is weakly better off.*
 - (b) *With narratives, the sender is strictly worse off iff $B < \alpha D_{\mathcal{M}'}$, $\omega_{\mathcal{M}}^{**} \geq c_{\mathcal{M}}$, $\omega_{\mathcal{M}'}^{**} \geq c_{\mathcal{M}'}$, and $\mathcal{M}$ satisfies the feasible-pool condition but $\mathcal{M}'$ does not.*
- (ii) *Suppose $\mathcal{M}'$ is more upward flexible than $\mathcal{M}$.*
 - (a) *Without narratives, the sender is strictly worse off iff $c_{\mathcal{M}'} < \omega_{\mathcal{M}'}^* \leq \omega_{\mathcal{M}}^* \leq c_{\mathcal{M}}$.*
 - (b) *With narratives, the sender is strictly worse off iff $B < \alpha D_{\mathcal{M}'}$ and either $c_{\mathcal{M}'} \leq \omega_{\mathcal{M}'}^{**} \leq \omega_{\mathcal{M}}^{**} < c_{\mathcal{M}}$, or $\omega_{\mathcal{M}}^{**} \geq c_{\mathcal{M}}$, $\omega_{\mathcal{M}'}^{**} \geq c_{\mathcal{M}'}$, and $\mathcal{M}$ satisfies the feasible-pool condition but $\mathcal{M}'$ does not.*

With narratives, greater flexibility can erode credibility through two channels. First, as without narratives, upward flexibility may destroy the credible separation needed for strategic messaging, but downward flexibility cannot. Nevertheless, narratives delay and cushion the loss: the lower cutoff $\omega_{\mathcal{M}}^{**} < \omega_{\mathcal{M}}^*$ allows complementarity to survive after pure messaging fails, and narrative persuasion alone may preserve the payoff p^*/α afterward ([Fig. 5](#)). Second, either direction of flexibility may enlarge message pools enough to violate the feasible-pool condition, making narrative persuasion alone infeasible. This is the only way downward

flexibility can hurt, so it weakly benefits the sender whenever the feasible-pool condition remains satisfied. This observation points to the role of downward flexibility in technologies that maximize the sender's persuasiveness across environments, which we characterize next.

5.2 Maximally Persuasive Technologies

The preceding results imply that under any information technology the sender's equilibrium payoff is at most $\min\{p^*(1 + 1/\alpha), 1\}$. We characterize the technologies that attain this bound uniformly across environments.

Definition 5. A technology $\mathcal{M} \in \mathbb{M}$ is *maximally persuasive* if its sender-optimal equilibrium payoff equals $\min\{p^*(1 + 1/\alpha), 1\}$ for every admissible environment (F, ω_0, α) .

Theorem 4 (Maximally Persuasive Technologies). *An information technology $\mathcal{M} = (L, U) \in \mathbb{M}$ is maximally persuasive if and only if:*

- (i) *any two types $\omega < \omega'$ in Ω share a feasible message; and*
- (ii) *every interior type is credible, i.e., $U(\omega) > U(\omega')$ for every $\omega \in \text{int}(\Omega)$ and $\omega' < \omega$.*

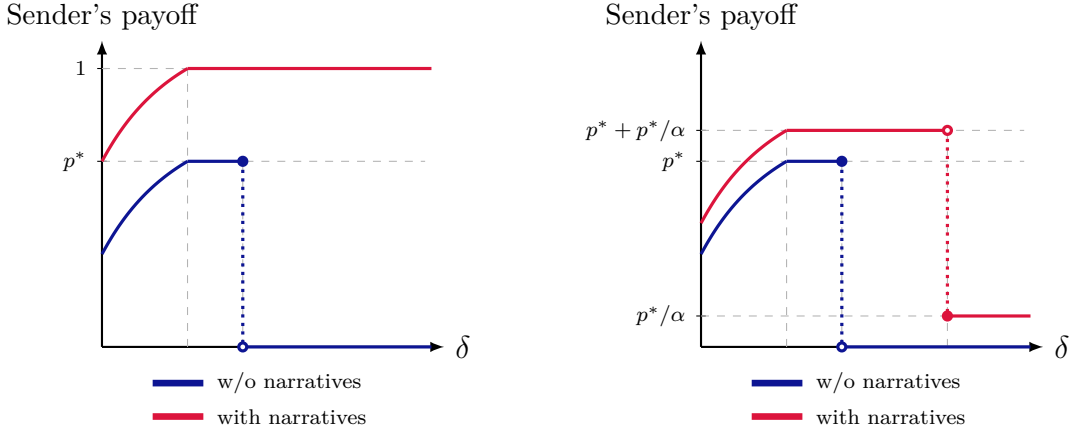
The two conditions resolve the flexibility–credibility tradeoff.³⁰ Condition (i) provides enough flexibility to attain the unconstrained BP payoff, $p_{\mathcal{M}}^* = p^*$, minimizing the narrative demand in every environment. Condition (ii) lets every interior cutoff type separate from lower types, preventing unraveling. One example is the understating technology $M(\omega) = \{\omega' \in \Omega : \omega' \leq \omega\}$. Any two types can pool at the lower type, while every interior type can separate from lower types by reporting truthfully. By comparison, cheap talk permits unrestricted pooling but lacks credible separation, whereas mandatory disclosure ensures separation but rules out pooling across types.

6 Extensions

We explore several extensions of the baseline model, focusing on cheap talk for tractability. Additional extensions and formal results appear in Supplemental Appendix B.4.

²⁹Both panels assume $F = \text{Unif}[0, 1]$, $\omega_0 = 2/3$, $L(\omega) = \omega$, and $U(\omega) = \min\{\delta\omega, 1\}$. Panel (a) sets $\alpha = 2$, while panel (b) sets $\alpha = 6$. In panel (a), the payoff with narratives happens to begin at p^* because $F(\omega \geq \omega_0) + p^*/\alpha = 1/3 + 1/3 = p^*$ under these parameters; this equality is coincidental.

³⁰The same conditions characterize technologies that attain p^* in equilibrium without narratives in every admissible environment. Thus, although narratives can reverse technology rankings within an environment, uniform optimality requires the same balance of flexibility and credibility.



(a) Full manipulation attainable: $B \geq \alpha D$. (b) Full manipulation unattainable: $B < \alpha D$.

Figure 5: Sender's optimal equilibrium payoff changes as upward flexibility increases.²⁹

Bayesian Updating over Models Suppose the receiver averages over models rather than switching between them. We allow the sender to propose an arbitrary narrative and assume the receiver assigns it prior probability $\beta_0 \in (0, 1)$ and the initial correctly specified model $1 - \beta_0$. After observing the message, he updates by Bayes' rule and acts on the model-averaged posterior. Our mechanism is robust here: full manipulation remains feasible if and only if β_0 is at least a cutoff $\bar{\beta}_0 \in (0, 1]$ (Proposition B.3). As in the baseline, the sender designs the narrative to reinterpret bad news as evidence of favorable states. Unlike switching, Bayesian averaging retains weight on the correctly specified model, so the receiver cannot be “misspecified” in the sense of entirely ruling out the true state.³¹ Manipulation therefore requires the narrative's suggested gains to offset the initial model's implied losses. Nevertheless, sufficient upside in favorable states can sustain manipulation even if the receiver initially favors his own model ($\beta_0 < \frac{1}{2}$).

An alternative approach fixes the receiver's initial models and prior, restricting the sender to proposing one of those models. The scope for persuasion then depends on the available alternatives. For example, if the only alternative is a babbling model, its posterior remains at the prior and cannot make a losing message persuasive. This underscores the importance of the sender's freedom to construct narratives in sustaining manipulation.

Multiple Narratives Suppose the sender can propose K narratives. The receiver adopts the one with the highest Bayes factor if it reaches α , breaking ties in the sender's

³¹An implication is that full manipulation is strictly harder under Bayesian averaging when the two rules are calibrated to the same model weights. Setting $\alpha = (1 - \beta_0)/\beta_0$ makes switching select the more probable model, requiring only $\beta_0 \geq D$ for full manipulation, whereas averaging requires $\beta_0 \geq \bar{\beta}_0 > D$.

favor.³² With sufficiently many messages, each narrative makes a different losing message persuasive, expanding the narrative budget to KB . Full manipulation is feasible if and only if $KB \geq \alpha D$ (Proposition B.4).

Narratives About the State Distribution If the narrative can also specify a prior over states, the sender can claim that only favorable states occur and a single message is always sent, raising the budget to one. She uses this message only for unfavorable states. Full manipulation is feasible if and only if $1 \geq \alpha D$ (Proposition B.5).

Initially Misspecified Models Suppose the receiver’s initial model may be incorrect and he evaluates the narrative against it rather than the sender’s actual strategy. If some message induces the risky action under the initial model, the sender achieves full manipulation by sending it in every state. Otherwise, full manipulation is feasible if and only if the initial model assigns probability at most p^*/α to some message (Proposition B.6). The sender sends this message and proposes a narrative that makes it good news, inducing a switch. Such a message always exists when the message space is infinite.

7 Concluding Remarks

Disinformation rampantly undermines political processes, economic outcomes, and public discourse in societies worldwide. While one would hope that informed, skeptical audiences who understand the speaker’s motives would be immune, our results show that even they can be systematically manipulated, potentially with certainty. By choosing what information to provide and proposing how to interpret it, the sender can fully manipulate a receiver who begins with a correct understanding of the world, exploiting the very openness to alternative viewpoints that would be reasonable in good-faith social interactions.

Beyond the finance example in the introduction, the manipulation tactics identified in our analysis arise across economic and social settings. One is *recommendation inflation*, where the sender claims to reserve favorable assessments for relatively good cases but inflates assessments across the board. A politician may portray failures as successes and genuine successes as exceptional achievements (Martínez, 2022); a selective university may award A’s to weaker students and A+’s to excellent ones (Denning et al., 2022). Another tactic is *interpretation inversion*, where the sender proposes a narrative that reverses the interpretation of the evidence. As a form of “gaslighting,”³³ an abusive partner may claim

³²When several Bayes factors are infinite, the receiver first selects a model with the highest likelihood of the observed message and breaks any remaining tie in the sender’s favor.

³³The [APA Dictionary of Psychology](#) defines gaslighting as “manipulat[ing] another person into doubting

that harshness signals love and kindness its absence (Sweet, 2019); a politician may claim that the mainstream media systematically inverts the truth (Schiff et al., 2025); a state propagandist can claim that incriminating images were staged by an adversary (Alyukov and Zavadskaya, 2026); a conspiracy theorist may claim that government rebuttals are part of a cover-up and thus evidence of the alleged conspiracy (Sunstein and Vermeule, 2009). These examples highlight how manipulation can operate without suppressing or altering evidence, simply by “claiming that black is white, in contradiction of the plain facts” (Orwell, 1949).

Our analysis shows that manipulation is troublingly robust but also identifies when it succeeds and when it fails. These conditions suggest potential targets for policy design, such as receiver skepticism, the scope of the sender’s narrative control, and the hardness of information.³⁴ Yet our results also suggest limits to familiar countermeasures. Fact-checking focused on information accuracy may leave misleading narratives intact; media literacy campaigns emphasizing strategic awareness may not suffice either, since our receiver already understands the sender’s incentives yet remains vulnerable. Promising directions for future work include designing effective policy interventions such as introducing reputation concerns in repeated interactions or leveraging competition among multiple senders. Understanding how people can be made to believe absurdities is the necessary first step toward safeguarding them from being made to act against their own interests and even commit injustices.

References

- ACEMOĞLU, D., C. G. AKSOY, C. BAYSAN, C. MOLINA, AND G. ZEKI (2025): “Misperceptions and Demand for Democracy under Authoritarianism,” Working Paper.
- AGARWAL, S., G. ATONDO-SIU, M. ORDEKIAN, A. HUTCHINGS, E. MARICONTI, AND M. VASEK (2023): “Short Paper: DeFi Deception—Uncovering the Prevalence of Rugpulls in Cryptocurrency Projects,” in *International Conference on Financial Cryptography and Data Security*, Springer, 363–372.
- AINA, C. (2025): “Tailored Stories,” Working Paper.
- AINA, C. AND F. H. SCHNEIDER (2025): “Weighting Competing Models,” Working Paper.

their perceptions, experiences, or understanding of events.”

³⁴Acemoğlu et al. (2025) provide experimental evidence suggesting that information and its interpretation both matter for correcting misperceptions about democratic institutions, and call for theory to understand these patterns. Our framework formalizes the joint role of information and interpretation, but also shows how a strategic sender can exploit both to create misperceptions.

- AKERLOF, G. A. AND R. J. SHILLER (2015): *Phishing for Phools: The Economics of Manipulation and Deception*, Princeton University Press.
- ALYUKOV, M. AND M. ZAVADSKAYA (2026): “‘It’s Not That Simple, We Don’t Know the Whole Truth’: The Effects of Pseudo Fact Checking in Wartime Russia,” *Perspectives on Politics*, 1–25.
- ANTIĆ, N. AND A. CHAKRABORTY (2026): “Selected Facts,” Working Paper.
- BA, C. (2026): “Robust Misspecified Models,” *American Economic Review*, 116, 1340–1379.
- BARRON, K. AND T. FRIES (2025): “Narrative Persuasion,” WZB Discussion Paper SP II 2023-301r, Wissenschaftszentrum Berlin für Sozialforschung (WZB), Berlin.
- BAUCH, G. AND M. FOERSTER (2024): “Strategic Communication of Narratives,” Working Paper.
- BOHREN, J. A. AND D. N. HAUSER (2021): “Learning with Heterogeneous Misspecified Models: Characterization and Robustness,” *Econometrica*, 89, 3025–3077.
- CRAWFORD, V. P. AND J. SOBEL (1982): “Strategic Information Transmission,” *Econometrica*, 50, 1431–1451.
- DE CLIPPEL, G. AND X. ZHANG (2022): “Non-Bayesian Persuasion,” *Journal of Political Economy*, 130, 2594–2642.
- DEKEL, E., B. L. LIPMAN, AND A. RUSTICHINI (1998): “Standard State-Space Models Preclude Unawareness,” *Econometrica*, 66, 159–173.
- DENNING, J. T., E. R. EIDE, K. J. MUMFORD, R. W. PATTERSON, AND M. WARNICK (2022): “Why Have College Completion Rates Increased?” *American Economic Journal: Applied Economics*, 14, 1–29.
- DHAWAN, A. AND T. J. PUTNIŃŠ (2023): “A New Wolf in Town? Pump-and-Dump Manipulation in Cryptocurrency Markets,” *Review of Finance*, 27, 935–975.
- ELIAZ, K. AND R. SPIEGLER (2026): “News Media as Suppliers of Narratives (and Information),” *Theoretical Economics*, 21, 928–972.
- ELIAZ, K., R. SPIEGLER, AND H. C. THYSEN (2021): “Strategic Interpretations,” *Journal of Economic Theory*, 192, 105192.
- FRICK, M., R. IJIMA, AND Y. ISHII (2023): “Belief Convergence under Misspecified Learning: A Martingale Approach,” *The Review of Economic Studies*, 90, 781–814.
- FRIESTAD, M. AND P. WRIGHT (1994): “The Persuasion Knowledge Model: How People

- Cope with Persuasion Attempts,” *Journal of Consumer Research*, 21, 1–31.
- GALPERTI, S. (2019): “Persuasion: The Art of Changing Worldviews,” *American Economic Review*, 109, 996–1031.
- GENNAIOLI, N. AND A. SHLEIFER (2010): “What Comes to Mind,” *The Quarterly Journal of Economics*, 125, 1399–1433.
- GROSSMAN, S. J. (1981): “The Informational Role of Warranties and Private Disclosure About Product Quality,” *The Journal of Law and Economics*, 24, 461–483.
- HART, S., I. KREMER, AND M. PERRY (2017): “Evidence Games: Truth and Commitment,” *American Economic Review*, 107, 690–713.
- HEIDHUES, P., B. KŐSZEGI, AND P. STRACK (2018): “Unrealistic Expectations and Misguided Learning,” *Econometrica*, 86, 1159–1214.
- ICHIHASHI, S. AND D. MENG (2021): “The Design and Interpretation of Information,” Working Paper.
- ISAAC, M. S. AND B. J. CALDER (2025): “Thirty Years of Persuasion Knowledge Research: From Demonstrating Effects to Building Theory to Increasing Applicability,” *Consumer Psychology Review*, 8, 3–14.
- JAIN, A. (2023): “Informing Agents Amidst Biased Narratives,” Working Paper.
- KAMENICA, E. AND M. GENTZKOW (2011): “Bayesian Persuasion,” *American Economic Review*, 101, 2590–2615.
- KO, H. (2025): “Persuasion in the Long Run: When History Matters,” Working Paper.
- LEUZ, C., S. MEYER, M. MUHN, E. SOLTES, AND A. HACKETHAL (2026): “Who Falls Prey to the Wolf of Wall Street? Investor Participation in Market Manipulation,” *Management Science*, 72, 6603–6631.
- LIPNOWSKI, E., D. RAVID, AND D. SHISHKIN (2022): “Persuasion via Weak Institutions,” *Journal of Political Economy*, 130, 2705–2730.
- LIZZERI, A., Y. LOU, AND J. PEREGO (2026): “The Limits of Verifiability: Credibility and Flexibility in Communication,” Working Paper.
- MALMENDIER, U. AND D. SHANTHIKUMAR (2007): “Are Small Investors Naive About Incentives?” *Journal of Financial Economics*, 85, 457–489.
- MARTÍNEZ, L. R. (2022): “How Much Should We Trust the Dictator’s GDP Growth Estimates?” *Journal of Political Economy*, 130, 2731–2769.
- MILGROM, P. R. (1981): “Good News and Bad News: Representation Theorems and

- Applications,” *The Bell Journal of Economics*, 12, 380–391.
- ORWELL, G. (1949): *Nineteen Eighty-Four*, Secker & Warburg.
- RAPPOPORT, D. (2025): “Evidence and Skepticism in Verifiable Disclosure Games,” *Theoretical Economics*, 20, 1213–1246.
- SCHIFF, K. J., D. S. SCHIFF, AND N. S. BUENO (2025): “The Liar’s Dividend: Can Politicians Claim Misinformation to Evade Accountability?” *American Political Science Review*, 119, 71–90.
- SCHWARTZSTEIN, J. AND A. SUNDERAM (2021): “Using Models to Persuade,” *American Economic Review*, 111, 276–323.
- (2024): “Sharing Models to Interpret Data,” Working Paper.
- SOBEL, J. (2020a): “Lying and Deception in Games,” *Journal of Political Economy*, 128, 907–947.
- (2020b): “Signaling Games,” in *Complex Social and Behavioral Systems: Game Theory and Agent-Based Models*, Springer, 251–268.
- STIGLITZ, J. E. AND A. KOSENKO (2024): “Robust Theory and Fragile Practice: Information in a World of Disinformation Part 2: Direct Communication,” in *The Elgar Companion to Information Economics*, ed. by D. R. Raban and J. Włodarczyk, Edward Elgar Publishing, 53–80.
- SUNSTEIN, C. R. AND A. VERMEULE (2009): “Conspiracy Theories: Causes and Cures,” *Journal of Political Philosophy*, 17, 202–227.
- SWEET, P. L. (2019): “The Sociology of Gaslighting,” *American Sociological Review*, 84, 851–875.
- TALEB, N. N. (2007): *The Black Swan: The Impact of the Highly Improbable*, New York: Random House.

A Proofs for Main Results

A.1 Off-path convention

Unless otherwise stated, our equilibrium constructions use the following convention at off-path histories, as defined in [Definition 1](#). Following any off-path narrative, the sender reports truthfully, and after every message the receiver retains the initial model and assigns the prior belief. Following the equilibrium narrative, impose the same receiver behavior at

off-path messages. In constructions without narratives, likewise assign the prior at off-path messages. These beliefs and model choices are unrestricted under weak PBE and induce the safe action. Thus truthful reporting is optimal following any off-path narrative, and neither an off-path narrative nor an off-path message offers the sender a profitable deviation.

A.2 Proofs for Section 3

A.2.1 Proof of Theorem 1

Proof. Full Manipulation. We first show that full manipulation is supported in equilibrium when $\alpha D \leq B$. Consider first an atomless state distribution, and let L, H, H' be three distinct messages in M . The sender proposes an optimal BP policy $\hat{\sigma}^* = \sigma_{BP}$, with $\sigma_{BP}(H|\omega) = 1$ for $\omega \geq \omega^*$ and $\sigma_{BP}(L|\omega) = 1$ otherwise. Following $\hat{\sigma}^*$, let $\sigma^*(H|\omega, \hat{\sigma}^*) = 1$ for $\omega < \omega^*$ and $\sigma^*(H'|\omega, \hat{\sigma}^*) = 1$ otherwise. Let $A^*(\pi) = \mathbb{1}\{\mathbb{E}_{\omega \sim \pi}[\omega] \geq \omega_0\}$. Following $\hat{\sigma}^*$, message H induces switching because $\lambda(H) = p^*/(1-p^*) = B/D \geq \alpha$, whereas H' induces retention of the initial model because $\lambda(H') = 0$. In both cases, the adopted model yields posterior $F(\cdot|\omega \geq \omega^*)$, with mean ω_0 , so the receiver takes the risky action. Following L , $\lambda(L) = +\infty$ induces switching, yielding posterior $F(\cdot|\omega < \omega^*)$ with mean below ω_0 , so he takes the safe action. Following $\hat{\sigma}^*$, every type obtains the maximal payoff of one, so no message deviation is profitable. Together with the off-path convention in Appendix A.1, this establishes a weak Narrative PBE supporting full manipulation.

For discrete states, let σ_{BP} send H for $\omega > \omega^*$, L for $\omega < \omega^*$, and H with probability q at ω^* , sending L otherwise. Here $q = \frac{\int_{(\omega^*, \sup \Omega]} (\omega - \omega_0) dF(\omega)}{(\omega_0 - \omega^*)F(\{\omega^*\})}$ if $\omega^* < \omega_0$, and $q = 1$ otherwise, so the posterior mean after H is ω_0 . Thus $B = F(\omega > \omega^*) + qF(\{\omega^*\})$ and $D = 1 - B$. Again propose $\hat{\sigma}^* = \sigma_{BP}$ and, following $\hat{\sigma}^*$, set $\sigma^*(H|\omega, \hat{\sigma}^*) = \sigma_{BP}(L|\omega)$ and $\sigma^*(H'|\omega, \hat{\sigma}^*) = \sigma_{BP}(H|\omega)$. The likelihood ratios remain $\lambda(H) = B/D$, $\lambda(H') = 0$, and $\lambda(L) = +\infty$. After either H or H' , the adopted posterior is the BP posterior following H ; after L , it is the BP posterior following L . The remaining equilibrium verification is unchanged.

No Manipulation. Suppose $\alpha D > B$. For any proposed model $\hat{\sigma}$ and actual strategy σ , let $W_1 := M_\sigma^+ \cap \{m : \lambda(m) < \alpha\}$ and $W_2 := M_\sigma^+ \cap \{m : \lambda(m) \geq \alpha\}$. Let $S_1 := P_\sigma(W_1)$ be the probability that the receiver retains the initial model and takes the risky action, and $S_2 := P_\sigma(W_2)$ the probability that he switches models and takes the risky action. First, $S_1 \leq P_\sigma(M_\sigma^+) = 1 - D(\sigma) \leq 1 - D$. Second, the switching rule gives $S_2 = P_\sigma(W_2) \leq P_{\hat{\sigma}}(W_2)/\alpha \leq B(\hat{\sigma})/\alpha \leq B/\alpha$. Hence $S_1 + S_2 \leq 1 - D + B/\alpha < 1$. Thus the sender loses with positive probability under every pair $(\hat{\sigma}, \sigma)$. If an equilibrium had positive winning probability, there would exist an on-path winning message and a positive

mass of types whose conditional winning probability is below one. Under cheap talk, each such type could deviate to that message and win with certainty, a contradiction. Hence the sender loses with probability one in equilibrium.

To establish existence, let $\hat{\sigma}^*(H|\omega) = \sigma^*(H|\omega, \hat{\sigma}^*) = 1$ for all ω . Either model yields the prior, inducing the safe action since $\mathbb{E}[\omega] < \omega_0$. Together with the off-path convention, every narrative and message yields zero, so no deviation is profitable. $\square$

A.2.2 Proof of Corollary 2

Proof. Consider the family $\{F_c\}_{c \in C}$ of probability distributions with $f_c(\omega) = f(\omega - c)$ with support $\Omega_c := \{\omega + c : \omega \in \Omega\}$. The admissible shift satisfies $c \in C := \{c \in \mathbb{R} : \mathbb{E}_{\omega \sim F_c}[\omega] < \omega_0 \text{ and } F_c(\omega \geq \omega_0) > 0\}$. To simplify notation, let $\mathbb{E}_{\omega \sim F_c}$ be denoted more succinctly as $\mathbb{E}_c$. Notice that $\mathbb{E}_c[\omega] = \mathbb{E}_0[\omega] + c$ and $\mathbb{E}_c[\omega|\omega \geq z] = \mathbb{E}_0[\omega|\omega \geq z - c] + c$, $\forall c \in C$ and $z \in \Omega_c$. The condition $\mathbb{E}_c[\omega] < \omega_0$ requires $c < \omega_0 - \mathbb{E}_0[\omega]$. The condition $F_c(\omega \geq \omega_0) > 0$ requires the upper endpoint of the shifted support to exceed ω_0 , or $\sup \Omega + c > \omega_0$. Combining these two restrictions gives $C = (\omega_0 - \sup \Omega, \omega_0 - \mathbb{E}_0[\omega])$.

For each $c \in C$, let ω_c^* solve $\mathbb{E}_c[\omega|\omega \geq \omega_c^*] = \omega_0$ and let $p_c^* := F_c([\omega_c^*, \infty))$. We now show p_c^* is continuous and strictly increasing in c . Let $G : (\inf \Omega, \sup \Omega) \rightarrow (\mathbb{E}_0[\omega], \sup \Omega)$ be given by $G(x) := \mathbb{E}_0[\omega|\omega \geq x]$. Since f is continuous and strictly positive on Ω , G is continuous and strictly increasing, and hence has a continuous inverse G^{-1} on $(\mathbb{E}_0[\omega], \sup \Omega)$. Recall that ω_c^* is defined to satisfy $G(\omega_c^* - c) = \mathbb{E}_0[\omega|\omega \geq \omega_c^* - c] = \mathbb{E}_c[\omega|\omega \geq \omega_c^*] - c = \omega_0 - c$, which can be rearranged as $\omega_c^* - c = G^{-1}(\omega_0 - c)$. It then follows that $\omega_c^* - c$ is continuous and strictly decreasing in c , so $p_c^* \equiv F_c(\omega \geq \omega_c^*) = F_0(\omega \geq \omega_c^* - c)$ is continuous and strictly increasing in c . Notice that as c approaches the lower end point of C , we have $\omega_0 - c$ approaching $\sup \Omega$, so $\omega_c^* - c = G^{-1}(\omega_0 - c) \rightarrow \sup \Omega$ and $p_c^* = F_0(\omega \geq \omega_c^* - c) \rightarrow 0$. Similarly, as c approaches the upper end point of C , $\omega_0 - c$ approaches $\mathbb{E}_0[\omega]$, so $\omega_c^* - c = G^{-1}(\omega_0 - c) \rightarrow \inf \Omega$ and $p_c^* \rightarrow 1$.

Hence, p_c^* is continuous and strictly increasing from 0 to 1 on C . Since $\frac{\alpha}{1+\alpha} \in (0, 1)$, there exists a unique $c^* \in C$ s.t. $p_{c^*}^* = \frac{\alpha}{1+\alpha}$ and $p_c^* \geq \frac{\alpha}{1+\alpha}$ if and only if $c \geq c^*$. By Theorem 1, if $c < c^*$, the sender cannot manipulate the receiver, so their payoffs are 0 and ω_0 , respectively; if $c \geq c^*$, the sender fully manipulates the receiver, so their payoffs are 1 and $\mathbb{E}_c[\omega] = \mathbb{E}_0[\omega] + c$, respectively. Thus, at $c = c^*$, the receiver's payoff drops discontinuously from ω_0 to $\mathbb{E}_{c^*}[\omega] < \omega_0$. For all larger admissible $c \in (c^*, \infty) \cap C$, the receiver's payoff increases linearly in c while remaining strictly below ω_0 . $\square$

A.3 Proofs for Sections 4 and 5

A.3.1 Proof of Lemma 2

Proof. Part (i). We first show that $\Omega_{\mathcal{M}}(m)$ is a closed interval, possibly unbounded. Let $\omega_l := \inf\{\omega \in \Omega : U(\omega) \geq m\}$ and $\omega_r := \sup\{\omega \in \Omega : L(\omega) \leq m\}$. Since type m can report truthfully, $\omega_l \leq m \leq \omega_r$. For any $\omega \in (\omega_l, \omega_r)$, the definitions of ω_l and ω_r imply that there exist $\omega' < \omega < \omega''$ in Ω such that $U(\omega') \geq m$ and $L(\omega'') \leq m$. Monotonicity then gives $L(\omega) \leq L(\omega'') \leq m \leq U(\omega') \leq U(\omega)$, so $\omega \in \Omega_{\mathcal{M}}(m)$.

It remains to check the finite endpoints, which belong to Ω because Ω is closed. If ω_l is finite, right-continuity of U gives $U(\omega_l) \geq m$, while $L(\omega_l) \leq \omega_l \leq m$, so $\omega_l \in \Omega_{\mathcal{M}}(m)$. Similarly, if ω_r is finite, left-continuity of L gives $L(\omega_r) \leq m$, while $U(\omega_r) \geq \omega_r \geq m$, so $\omega_r \in \Omega_{\mathcal{M}}(m)$. No type below ω_l or above ω_r can send m , by definition. Hence $\Omega_{\mathcal{M}}(m)$ is a closed interval, unbounded below if $\omega_l = -\infty$ and unbounded above if $\omega_r = +\infty$.

Part (ii). Suppose $\omega \in [\omega_1, \omega_2] \subseteq \Omega_{\mathcal{M}}(m)$ can send m' . Monotonicity gives $L(\omega_1) \leq L(\omega) \leq m' \leq U(\omega) \leq U(\omega_2)$. If $m' \leq U(\omega_1)$, then $m' \in M(\omega_1)$. Otherwise, since ω_1 and ω_2 can both send m , we have $m' > U(\omega_1) \geq m \geq L(\omega_2)$, so $m' \in M(\omega_2)$. $\square$

A.3.2 Proof of Proposition 1

Proof. The proof proceeds in three steps: we construct a candidate cutoff strategy, show that no feasible cutoff strategy yields a higher payoff, and establish that every optimal strategy has a cutoff structure.

Step 1. Let $\bar{U}_0 := \sup_{\omega < \omega_0} U(\omega)$. For a candidate cutoff $\tilde{\omega} \in \Omega$ with $\tilde{\omega} < \omega_0$, we construct a *saving strategy* $\sigma_{\tilde{\omega}} : [\omega_0, \bar{\omega}(\tilde{\omega})) \rightarrow [U(\tilde{\omega}), \bar{U}_0]$, where $\bar{\omega}(\tilde{\omega})$ is defined below. Low types in $[\tilde{\omega}, \omega_0)$ send their highest feasible messages, while high types are assigned to these messages in increasing order so that their surplus offsets the low types' deficit relative to ω_0 . Define the cumulative low-type deficit $\mathcal{L}_{\tilde{\omega}}(x) := \int_{\tilde{\omega}}^x (\omega_0 - t) f(t) dt$ for $x \in [\tilde{\omega}, \omega_0]$, and the cumulative high-type surplus $\mathcal{H}(y) := \int_{\omega_0}^y (t - \omega_0) f(t) dt$ for $y \in \Omega$ with $y \geq \omega_0$. Both functions are continuous and strictly increasing. If $\mathcal{L}_{\tilde{\omega}}(\omega_0) > \mathcal{H}(\sup \Omega)$, the available surplus cannot cover the deficit, and we declare $\sigma_{\tilde{\omega}}$ undefined and the cutoff infeasible. Otherwise, define $\bar{\omega}(\tilde{\omega})$ by $\mathcal{H}(\bar{\omega}(\tilde{\omega})) = \mathcal{L}_{\tilde{\omega}}(\omega_0)$, allowing $\bar{\omega}(\tilde{\omega}) = \sup \Omega$ and interpreting the equality as a limit when this endpoint is infinite. For $y \in [\omega_0, \bar{\omega}(\tilde{\omega}))$, set $x_{\tilde{\omega}}(y) := \mathcal{L}_{\tilde{\omega}}^{-1}(\mathcal{H}(y))$ and $\sigma_{\tilde{\omega}}(y) := U(x_{\tilde{\omega}}(y))$. Thus each high type sends the same message as its matched low type.

We now verify that each constructed pool of messages used above has posterior mean ω_0 , almost surely. Differentiating the matching identity $\mathcal{H}(y) = \mathcal{L}_{\tilde{\omega}}(x_{\tilde{\omega}}(y))$ gives $(y - \omega_0)f(y) = (\omega_0 - x_{\tilde{\omega}}(y))f(x_{\tilde{\omega}}(y))x'_{\tilde{\omega}}(y)$. Hence, changing variables from y to $x = x_{\tilde{\omega}}(y)$,

for every Borel set of messages A ,

$$\int_{\omega_0}^{\bar{\omega}(\tilde{\omega})} \mathbb{1}\{\sigma_{\tilde{\omega}}(y) \in A\} (y - \omega_0) f(y) dy = \int_{\tilde{\omega}}^{\omega_0} \mathbb{1}\{U(x) \in A\} (\omega_0 - x) f(x) dx.$$

Thus, among the types included in the construction, the posterior mean equals ω_0 for almost every message under the induced distribution.

We next check feasibility of the saving strategy. Since $x_{\tilde{\omega}}(y) < \omega_0 \leq y$, monotonicity gives $\sigma_{\tilde{\omega}}(y) = U(x_{\tilde{\omega}}(y)) \leq U(y)$. It therefore remains to check $\sigma_{\tilde{\omega}}(y) \geq L(y)$. At $\tilde{\omega} = \omega_0$, no pooling is required; set $\bar{\omega}(\omega_0) = \omega_0$, then feasibility holds trivially. Thus the set $\mathcal{F} := \{\tilde{\omega} \in \Omega : \tilde{\omega} \leq \omega_0, \sigma_{\tilde{\omega}} \text{ is well-defined and } \sigma_{\tilde{\omega}}(y) \geq L(y) \forall y \in [\omega_0, \bar{\omega}(\tilde{\omega})]\}$ is nonempty. Define $\omega_{\mathcal{M}}^* := \inf \mathcal{F}$. Every $\tilde{\omega} \in \mathcal{F}$ satisfies $\mathcal{L}_{\tilde{\omega}}(\omega_0) \leq \mathcal{H}(\sup \Omega)$, equivalently $\int_{\tilde{\omega}}^{\sup \Omega} (t - \omega_0) f(t) dt \geq 0$. This integral is zero at the unconstrained BP cutoff ω^* ; lowering the cutoff adds only types below ω_0 , making it strictly negative. Hence every feasible cutoff is at least ω^* , and $\omega_0 \in \mathcal{F}$ implies $\omega_{\mathcal{M}}^* \in [\omega^*, \omega_0] \subseteq \Omega$.

We now show that the infimum is attained. Suppose $\omega_{\mathcal{M}}^* \notin \mathcal{F}$, and choose $\tilde{\omega}_n \downarrow \omega_{\mathcal{M}}^*$ with $\tilde{\omega}_n \in \mathcal{F}$. Continuity gives $\mathcal{L}_{\omega_{\mathcal{M}}^*}(\omega_0) = \lim_n \mathcal{L}_{\tilde{\omega}_n}(\omega_0) \leq \mathcal{H}(\sup \Omega)$, so $\sigma_{\omega_{\mathcal{M}}^*}$ is well-defined. Fix $y \in [\omega_0, \bar{\omega}(\omega_{\mathcal{M}}^*)]$. For sufficiently large n , $\mathcal{H}(y) < \mathcal{L}_{\tilde{\omega}_n}(\omega_0)$, so $x_{\tilde{\omega}_n}(y)$ is defined. Splitting the deficit integral at $\tilde{\omega}_n$ and using the matching identity gives

$$\mathcal{L}_{\omega_{\mathcal{M}}^*}(x_{\tilde{\omega}_n}(y)) = \int_{\omega_{\mathcal{M}}^*}^{\tilde{\omega}_n} (\omega_0 - t) dF(t) + \int_{\tilde{\omega}_n}^{x_{\tilde{\omega}_n}(y)} (\omega_0 - t) dF(t) = \int_{\omega_{\mathcal{M}}^*}^{\tilde{\omega}_n} (\omega_0 - t) dF(t) + \mathcal{H}(y).$$

The first term decreases to zero. Since $\mathcal{L}_{\omega_{\mathcal{M}}^*}$ has a continuous, strictly increasing inverse, it follows that $x_{\tilde{\omega}_n}(y) \downarrow x_{\omega_{\mathcal{M}}^*}(y)$. Right-continuity of U then gives $\sigma_{\omega_{\mathcal{M}}^*}(y) = \lim_n U(x_{\tilde{\omega}_n}(y)) \geq L(y)$, since each $\tilde{\omega}_n$ is feasible. Thus $\omega_{\mathcal{M}}^* \in \mathcal{F}$.

Let $\bar{\omega} := \bar{\omega}(\omega_{\mathcal{M}}^*)$. Given $\omega_{\mathcal{M}}^*$, define the candidate messaging strategy σ^* by: (i) $\sigma^*(\omega) = \omega$ for all $\omega < \omega_{\mathcal{M}}^*$, (ii) $\sigma^*(\omega) = U(\omega)$ for all $\omega \in [\omega_{\mathcal{M}}^*, \omega_0]$, (iii) $\sigma^*(\omega) = \sigma_{\omega_{\mathcal{M}}^*}(\omega)$ for all $\omega \in [\omega_0, \bar{\omega})$, and (iv) $\sigma^*(\omega) = U(\omega)$ for all $\omega \geq \bar{\omega}$. By construction σ^* is feasible and almost all types $\omega \geq \omega_{\mathcal{M}}^*$ win under σ^* .

Step 2. Suppose, toward a contradiction, that some feasible $\sigma' \in \Sigma$ makes almost all types $\omega \geq \tilde{\omega}$ win for a cutoff $\tilde{\omega} < \omega_{\mathcal{M}}^*$. Then $\sigma_{\tilde{\omega}}$ is either not well-defined or violates the lower bound L . We derive a contradiction in each case.

Case A: $\sigma_{\tilde{\omega}}$ is not well-defined. The total low-type deficit exceeds the available high-type surplus. By continuity, there exists $\hat{\omega} \in (\tilde{\omega}, \omega_0)$ such that $\mathcal{H}(\sup \Omega) < \mathcal{L}_{\tilde{\omega}}(\hat{\omega})$. But the surplus available from high types able to send messages weakly below $U(\hat{\omega})$ is at most $S' := \int_{\{\omega \geq \omega_0 : L(\omega) \leq U(\hat{\omega})\}} (\omega - \omega_0) f(\omega) d\omega \leq \mathcal{H}(\sup \Omega) < \mathcal{L}_{\tilde{\omega}}(\hat{\omega})$. For σ' to make almost all types in $[\tilde{\omega}, \hat{\omega}]$ win, these types must send some message weakly below $U(\hat{\omega})$. Integrating

over all $\omega \in [\tilde{\omega}, \hat{\omega}]$ requires total surplus at least $\mathcal{L}_{\tilde{\omega}}(\hat{\omega})$ from high types, but the available surplus is only $\mathcal{S}' < \mathcal{L}_{\tilde{\omega}}(\hat{\omega})$ —a contradiction.

Case B: $\sigma_{\tilde{\omega}}$ is well-defined but violates the lower bound. There exists $\omega_h \in [\omega_0, \bar{\omega}(\tilde{\omega})]$ such that, writing $\hat{\omega} := x_{\tilde{\omega}}(\omega_h)$, we have $\sigma_{\tilde{\omega}}(\omega_h) = U(\hat{\omega}) < L(\omega_h)$. We may take $\omega_h > \omega_0$, since right-continuity of $\sigma_{\tilde{\omega}}$ and monotonicity of L ensure that any strict violation at ω_0 persists immediately above it. Monotonicity gives $L(\omega) \geq L(\omega_h) > U(\hat{\omega})$ for every $\omega \geq \omega_h$, so none of these types can send a message weakly below $U(\hat{\omega})$.

Every type in $[\tilde{\omega}, \hat{\omega}]$ can send only messages weakly below $U(\hat{\omega})$. Making almost all these types win therefore requires high-type surplus at least $\mathcal{L}_{\tilde{\omega}}(\hat{\omega}) = \mathcal{H}(\omega_h)$ from types able to send such messages. But the available surplus satisfies $\mathcal{S}' := \int_{\{\omega \geq \omega_0: L(\omega) \leq U(\hat{\omega})\}} (\omega - \omega_0) f(\omega) d\omega < \mathcal{H}(\omega_h) = \mathcal{L}_{\tilde{\omega}}(\hat{\omega})$, a contradiction. The inequality is strict since left-continuity of L excludes an interval of high types immediately below ω_h .

So no feasible strategy makes almost all types above a cutoff below $\omega_{\mathcal{M}}^*$ win, and σ^* is optimal among strategies whose winning region is an upper interval up to null sets.

Step 3. Finally, we show that every optimal strategy has the cutoff structure. Consider any feasible strategy σ' , and let $q(\omega) \in [0, 1]$ denote type ω 's winning probability. Define $K := \int_{\inf \Omega}^{\omega_0} q(t) f(t) dt$ and $\mathcal{L}_q(x) := \int_{\inf \Omega}^x (\omega_0 - t) q(t) f(t) dt$ for $x \leq \omega_0$. Thus K is the winning mass of low types, and $\mathcal{L}_q(x)$ is the deficit contributed by winning low types weakly below x . As in Step 2, these types can send only messages weakly below $U(x)$. Integrating the receiver's posterior-mean condition over winning messages in this range gives $\mathcal{L}_q(x) \leq \int_{\{t \geq \omega_0: L(t) \leq U(x)\}} (t - \omega_0) f(t) dt$ for $x < \omega_0$. Integrating over all winning messages also gives $\mathcal{L}_q(\omega_0) \leq \mathcal{H}(\sup \Omega)$.

Suppose first that $K > 0$, and choose $c < \omega_0$ such that $F([c, \omega_0]) = K$. Such a cutoff exists: K cannot equal the entire low-type mass, since $\mathbb{E}[\omega] < \omega_0$ implies that its total deficit exceeds the available high-type surplus. For every $x \in [c, \omega_0]$, $\int_{\inf \Omega}^x q(t) f(t) dt = K - \int_x^{\omega_0} q(t) f(t) dt \geq K - F((x, \omega_0)) = F([c, x])$. Among types weakly below x , the interval $[c, x]$ supplies this mass with the smallest deficit, because $\omega_0 - t$ decreases with t . Hence $\mathcal{L}_c(x) \leq \mathcal{L}_q(x)$ for every $x \in [c, \omega_0]$.

These inequalities imply that the saving strategy at c is feasible. The aggregate surplus condition holds because $\mathcal{L}_c(\omega_0) \leq \mathcal{L}_q(\omega_0) \leq \mathcal{H}(\sup \Omega)$. If the lower feasibility bound failed, Case B of Step 2 would give some $\hat{\omega} \in [c, \omega_0]$ for which the available surplus at messages weakly below $U(\hat{\omega})$ is strictly less than $\mathcal{L}_c(\hat{\omega})$. This contradicts $\mathcal{L}_c(\hat{\omega}) \leq \mathcal{L}_q(\hat{\omega})$ and the surplus bound above. Thus $c \in \mathcal{F}$, so $c \geq \omega_{\mathcal{M}}^*$. The winning probability under σ' therefore satisfies $\int_{\Omega} q(t) f(t) dt \leq F([\omega_0, \sup \Omega]) + K = F([c, \sup \Omega]) \leq p_{\mathcal{M}}^*$. If $K = 0$, the same

bound follows from $\omega_{\mathcal{M}}^* \leq \omega_0$. Since the candidate strategy attains $p_{\mathcal{M}}^*$, it is globally optimal.

For an optimal σ' with $K > 0$, both inequalities in the payoff bound must bind. Thus almost every high type wins and $c = \omega_{\mathcal{M}}^*$. This identifies the mass of winning low types, but not their locations. We now show that almost all winning low types lie in $[c, \omega_0]$. Suppose instead that a positive mass below c wins, so $\eta := \int_{\inf \Omega}^c (c-t)q(t)f(t) dt > 0$. For every $x \in [c, \omega_0]$,

$$\begin{aligned} \mathcal{L}_q(x) - \mathcal{L}_c(x) &= \eta + \int_c^x (t-c)(1-q(t))f(t) dt \\ &\quad + (\omega_0 - c) \left(\int_{\inf \Omega}^x q(t)f(t) dt - F([c, x]) \right) \geq \eta, \end{aligned}$$

where the last term is nonnegative by the preceding mass comparison.

Choose $a < c$ with $\mathcal{L}_q(a) > 0$, then choose $c' \in (a, c)$ sufficiently close to c that $\mathcal{L}_{c'}(c) < \min\{\eta, \mathcal{L}_q(a)\}$. For $x \in [c', c]$, $\mathcal{L}_{c'}(x) \leq \mathcal{L}_{c'}(c) < \mathcal{L}_q(a) \leq \mathcal{L}_q(x)$. For $x \in [c, \omega_0]$, $\mathcal{L}_{c'}(x) = \mathcal{L}_{c'}(c) + \mathcal{L}_c(x) < \eta + \mathcal{L}_c(x) \leq \mathcal{L}_q(x)$. The same feasibility argument therefore implies $c' \in \mathcal{F}$, contradicting $c = \omega_{\mathcal{M}}^* = \inf \mathcal{F}$.

Thus, almost no type below c wins. Since the winning low-type mass is $F([c, \omega_0])$, almost every type in that interval wins. If an optimum instead has $K = 0$, equality in the payoff bound implies $\omega_{\mathcal{M}}^* = \omega_0$ and almost every high type wins. In either case, every optimal strategy makes almost all types above $\omega_{\mathcal{M}}^*$ win and almost all types below it lose. $\square$

A.3.3 Proof of Proposition 2

Proof. We first establish an upper bound on the equilibrium winning probability, then construct an equilibrium attaining this bound, and finally show that all types lose in every equilibrium when $\tilde{\omega}_{\mathcal{M}}^*$ is undefined.

Step 1. We first show that winning types form an upper interval almost surely. Suppose this property fails on a set of positive probability. Choose types $\omega_1 < \omega_2$ such that ω_1 wins and ω_2 loses, and their respective messages m^+ and m^- at which Bayes' rule applies. The posterior at m^+ includes a high type $\omega_H > \omega_0$. If $\omega_2 \leq \omega_H$, Lemma 2(i) implies that ω_2 can also send m^+ and win, a contradiction. Otherwise, $\omega_2 > \omega_H$. Bayes' rule and sender optimality imply that the posterior at m^- includes a losing low type $\omega_L < \omega_0$. If $\omega_L \geq \omega_1$, the same interval property lets ω_L send m^+ . If $\omega_L < \omega_1$, both ω_L and ω_2 can send m^- , while the intermediate type ω_1 can send m^+ . By Lemma 2(ii), at least one of the two losing types can send m^+ and profitably deviate, again a contradiction.

Therefore, any equilibrium with positive winning probability has a cutoff $\hat{\omega}$: almost every type strictly above $\hat{\omega}$ wins, and almost every type strictly below it loses, with no

restriction on the cutoff type itself. By [Proposition 1](#), $\hat{\omega} \geq \omega_{\mathcal{M}}^*$. We now show that credible types must exist arbitrarily close to this cutoff from above. Consider any winning type $\omega > \hat{\omega}$ and a winning message it sends. The lower endpoint of the message's feasible-type interval lies in $[\hat{\omega}, \omega]$: an endpoint below $\hat{\omega}$ would allow a positive mass of losing types to deviate to that message. Moreover, this endpoint is credible, since right-continuity of U ensures that it can send the message, whereas every lower type has an upper bound strictly below the message. Taking winning types arbitrarily close to $\hat{\omega}$ therefore gives $\tilde{\omega}_{\mathcal{M}}^* \leq \hat{\omega}$, so the equilibrium winning probability is at most $\tilde{p}_{\mathcal{M}}^*$. If no credible type at or above $\omega_{\mathcal{M}}^*$ exists, the same argument rules out any equilibrium with positive winning probability.

Step 2. We construct an equilibrium attaining this bound.

Case 1: $\tilde{\omega}_{\mathcal{M}}^* \leq \omega_0$. Take the complete constrained-BP strategy constructed in [Proposition 1](#), denoted by σ_{BP} , and modify it as follows. If $\tilde{\omega}_{\mathcal{M}}^*$ is credible, any type that previously sent a message $m \geq U(\tilde{\omega}_{\mathcal{M}}^*)$ continues to send that message. If the cutoff is not credible, retain only messages $m > U(\tilde{\omega}_{\mathcal{M}}^*)$. All other types instead send $U(\omega)$. Call this modified strategy σ^* . It is feasible because each type either keeps its original feasible message or sends its upper bound.

When the cutoff is credible, $U(\omega) < U(\tilde{\omega}_{\mathcal{M}}^*)$ for every $\omega < \tilde{\omega}_{\mathcal{M}}^*$. When it is not credible, every $\omega > \tilde{\omega}_{\mathcal{M}}^*$ has a credible type strictly between itself and the cutoff (by the definition of $\tilde{\omega}_{\mathcal{M}}^*$ as an infimum of credible types), so $U(\omega) > U(\tilde{\omega}_{\mathcal{M}}^*)$. Thus low types strictly above the cutoff keep their messages, and types below it cannot access the retained messages or the upper-bound messages of types strictly above it. The retained pools remain winning, and any reassigned high types either join these pools or form new winning pools, as in the construction in [Proposition 1](#). All remaining messages are sent only by types weakly below $\tilde{\omega}_{\mathcal{M}}^* \leq \omega_0$ and lose almost surely. Thus the winning probability is $F(\omega \geq \tilde{\omega}_{\mathcal{M}}^*)$.

Case 2: $\tilde{\omega}_{\mathcal{M}}^* > \omega_0$, or no credible type exists at or above $\omega_{\mathcal{M}}^*$. Define $\sigma^*(\omega) = U(\omega)$ for every type and write $u_0 := U(\omega_{\mathcal{M}}^*)$. In either case, $\omega_{\mathcal{M}}^*$ is not credible, so some $\omega' < \omega_{\mathcal{M}}^*$ satisfies $U(\omega') = u_0$. Moreover, $U = u_0$ on $[\omega_{\mathcal{M}}^*, \tilde{\omega}_{\mathcal{M}}^*)$ when the cutoff exists, and on all types weakly above $\omega_{\mathcal{M}}^*$ otherwise. Indeed, if $U(\omega) > u_0$ anywhere in this range, right-continuity gives a first type whose upper bound reaches $U(\omega)$. That type would be credible, contradicting the absence of credible types in this range.

Messages below u_0 are sent only by types below $\omega_{\mathcal{M}}^* \leq \omega_0$ and lose. Messages above u_0 , if any, are sent only by types weakly above $\tilde{\omega}_{\mathcal{M}}^* > \omega_0$ and win. The common message u_0 has positive probability because every type in $[\omega', \omega_{\mathcal{M}}^*]$ sends it. It must lose, because otherwise, almost all types $\omega \geq \omega'$ would win, contradicting [Proposition 1](#) since $\omega' < \omega_{\mathcal{M}}^*$.

If $\tilde{\omega}_{\mathcal{M}}^*$ exists, every $\omega > \tilde{\omega}_{\mathcal{M}}^*$ has a credible type in $[\tilde{\omega}_{\mathcal{M}}^*, \omega)$, so $U(\omega) > u_0$ and it wins. Types below the cutoff lose, while the cutoff type wins exactly when its upper bound exceeds u_0 . Every losing type has an upper bound weakly below u_0 and therefore cannot access a winning message. If the cutoff is undefined, no type can send a message above u_0 , so all types lose. At on-path messages where Bayes' rule leaves beliefs unspecified, assign probability one to a type that sends the message: one weakly above ω_0 for a winning message and strictly below ω_0 for a losing message. Such types exist by construction, so these beliefs satisfy the posterior-support restriction. Together with the off-path convention, neither construction admits a profitable deviation, and both attain $\tilde{p}_{\mathcal{M}}^* \leq p_{\mathcal{M}}^*$.

Step 3. Suppose no credible type exists at or above $\omega_{\mathcal{M}}^*$. By Step 1, every equilibrium has zero winning probability. If some type still wins after m , the posterior-support restriction implies some $\omega_H \geq \omega_0$ belongs to $\Omega_{\mathcal{M}}(m)$. Since this set is an interval, it must equal $\{\omega_H\}$; otherwise, a positive mass of types could send m and win, a contradiction. For every $\omega < \omega_H$, monotonicity gives $L(\omega) \leq L(\omega_H) \leq m$, so infeasibility of m requires $U(\omega) < m \leq U(\omega_H)$. Thus $\omega_H \geq \omega_0 \geq \omega_{\mathcal{M}}^*$ is credible, a contradiction. So every type loses in every equilibrium. $\square$

A.3.4 Proof of Theorem 2

Proof. Sufficiency. Suppose $B \geq \alpha D_{\mathcal{M}}$. Let σ^* be the constrained BP solution from Proposition 1, in which losing types report truthfully. Its losing-message set $M_{\sigma^*}^-$ has probability $P_{\sigma^*}(M_{\sigma^*}^-) = D_{\mathcal{M}}$. Construct $\hat{\sigma}^*$ as follows. Every type $\omega \geq \omega^*$ randomizes over losing messages according to their relative probabilities under σ^* : for every measurable message set E , let $\hat{\sigma}^*(E|\omega) = P_{\sigma^*}(E \cap M_{\sigma^*}^-)/D_{\mathcal{M}}$. Every type $\omega < \omega^*$ instead sends a single message $m' \in M \setminus \Omega$, which exists by assumption and is unused by σ^* . This is admissible because proposed models are unconstrained.

Since $F(\omega \geq \omega^*) = B$, we have $P_{\hat{\sigma}^*}(E \cap M_{\sigma^*}^-) = (B/D_{\mathcal{M}})P_{\sigma^*}(E \cap M_{\sigma^*}^-)$ for every measurable E . Hence $\lambda(m) = B/D_{\mathcal{M}} \geq \alpha$ at almost every losing message under σ^* . The receiver switches and obtains posterior $F(\cdot|\omega \geq \omega^*)$, with mean ω_0 , because these types use the same proposed message distribution. At almost every BP-winning message, the proposed probability density is zero, so $\lambda(m) = 0$ and the receiver keeps the initial model and takes the risky action.

At remaining messages sent under σ^* where likelihood ratios or posteriors are undetermined, choose versions consistent with the posterior-support restriction that induce the risky action. At the proposed-only message m' , the receiver switches and adopts $F(\cdot|\omega < \omega^*)$,

inducing the safe action. Complete the construction using the off-path convention in [Appendix A.1](#). Every type obtains payoff 1, so no message or narrative deviation is profitable.

Necessity. Suppose $B < \alpha D_{\mathcal{M}}$. Fix any $\hat{\sigma} \in \bar{\Sigma}$ and feasible $\sigma \in \Sigma$, and define

$$S_1 := P_{\sigma}(M_{\sigma}^+ \cap \{\lambda < \alpha\}), \quad S_2 := P_{\sigma}(M_{\hat{\sigma}}^+ \cap \{\lambda \geq \alpha\}). \quad (14)$$

These are the probabilities that the receiver takes the risky action while retaining the initial model or switching, respectively. By [Proposition 1](#), $S_1 \leq P_{\sigma}(M_{\sigma}^+) \leq p_{\mathcal{M}}^* = 1 - D_{\mathcal{M}}$. On messages that induce switching, the proposed-model density is at least α times the actual density. Integrating over winning messages in this set gives $\alpha S_2 \leq P_{\hat{\sigma}}(M_{\hat{\sigma}}^+ \cap \{\lambda \geq \alpha\}) \leq B(\hat{\sigma}) \leq B$, where the last inequality follows from [Lemma 1](#). Thus $S_1 + S_2 \leq 1 - D_{\mathcal{M}} + B/\alpha < 1$, ruling out full manipulation. $\square$

A.3.5 Proof of [Theorem 3](#)

Proof. Throughout, use the off-path convention in [Appendix A.1](#). Let S_1 and S_2 denote the probabilities of inducing the risky action without and with switching, respectively, as defined in [Eq. \(14\)](#). The proof of [Theorem 2](#) establishes $S_1 \leq p_{\mathcal{M}}^*$ and $S_2 \leq B/\alpha$ for any feasible messaging strategy and proposed model. Hence $S_1 + S_2 \leq p_{\mathcal{M}}^* + B/\alpha = F(\omega \geq \omega_{\mathcal{M}}^{**})$, where the equality follows from the definition of $\omega_{\mathcal{M}}^{**}$.

Part (i): $\omega_{\mathcal{M}}^{**} < c_{\mathcal{M}}$. We construct an equilibrium in which types $\omega \geq \omega_{\mathcal{M}}^{**}$ win. Regularity implies that $\omega_{\mathcal{M}}^{**}$ is credible and $U(\omega_{\mathcal{M}}^{**}) < U(\omega_{\mathcal{M}}^*)$. Let types $\omega \geq \omega_{\mathcal{M}}^*$ follow the constrained BP strategy from [Proposition 1](#), types $\omega < \omega_{\mathcal{M}}^{**}$ report truthfully, and types $\omega \in [\omega_{\mathcal{M}}^{**}, \omega_{\mathcal{M}}^*)$ send $\max\{U(\omega_{\mathcal{M}}^{**}), L(\omega)\}$. The latter message is feasible because both terms are weakly below $U(\omega)$. Let $M^{\text{lose}} := \{m \in \Omega : m < \omega_{\mathcal{M}}^{**}\}$ and let M^- collect the messages sent by intermediate types. Since $M^- \subseteq [U(\omega_{\mathcal{M}}^{**}), U(\omega_{\mathcal{M}}^*)]$ and the retained BP messages are weakly above $U(\omega_{\mathcal{M}}^*)$, these three message sets are disjoint. Moreover, $P_{\sigma^*}(M^-) = D_{\mathcal{M}} - F(\omega < \omega_{\mathcal{M}}^{**}) = B/\alpha$. Under $\hat{\sigma}^*$, every type $\omega \geq \omega^*$ randomizes over M^- according to its relative probabilities under σ^* : for every measurable message set E , let $\hat{\sigma}^*(E|\omega) = P_{\sigma^*}(E \cap M^-)/P_{\sigma^*}(M^-)$. Every type $\omega < \omega^*$ instead sends a single message $m_0 \in M \setminus \Omega$. This is feasible since $\hat{\sigma}^* \in \bar{\Sigma}$ is unconstrained and M contains at least two messages outside Ω .

We verify the receiver's response to each group of messages. First, messages in M^- induce switching and the risky action. Since $F(\omega \geq \omega^*) = B$, the induced distribution satisfies $P_{\hat{\sigma}^*}(E \cap M^-) = \alpha P_{\sigma^*}(E \cap M^-)$, giving $\lambda(m) = \alpha$ almost surely on M^- . Because these types use the same message distribution under $\hat{\sigma}^*$, the receiver's posterior after switching is $F(\cdot|\omega \geq \omega^*)$, with mean ω_0 . Second, retained BP messages have $\lambda(m) = 0$ almost surely, so the receiver retains σ^* and its BP posteriors and takes the risky action.

Third, messages in M^{lose} also have $\lambda(m) = 0$ but reveal $\omega = m < \omega_{\mathcal{M}}^{**} < \omega_0$, inducing the safe action. Finally, the proposed-only message m_0 induces switching to $F(\cdot | \omega < \omega^*)$ and the safe action. Where likelihood ratios or posteriors are undetermined, choose versions consistent with these actions and the posterior-support restriction.

Thus every type $\omega \geq \omega_{\mathcal{M}}^{**}$ wins. No lower type can access a winning message, since credibility gives $U(\omega) < U(\omega_{\mathcal{M}}^{**})$ for every $\omega < \omega_{\mathcal{M}}^{**}$, while every winning message is weakly above $U(\omega_{\mathcal{M}}^{**})$. Together with the off-path convention, this rules out message and narrative deviations. The construction attains the upper bound $F(\omega \geq \omega_{\mathcal{M}}^{**}) = p_{\mathcal{M}}^* + B/\alpha$ and is therefore sender-optimal.

Part (ii): $\omega_{\mathcal{M}}^{**} \geq c_{\mathcal{M}}$. Let $\bar{\Omega}_{\mathcal{M}} := \{\omega \in \Omega : U(\omega) = \sup \Omega\}$. Here Ω is bounded above and $c_{\mathcal{M}} \leq \omega_{\mathcal{M}}^{**} < \omega_{\mathcal{M}}^* \leq \omega_0$. We first show that $S_1 = 0$ in every equilibrium.

For $\omega' < \omega$ in $\bar{\Omega}_{\mathcal{M}}$, monotonicity of L gives $M(\omega) \subseteq M(\omega')$. Thus, if a type in $\bar{\Omega}_{\mathcal{M}}$ loses, every higher type also loses; otherwise, the losing type could deviate to a higher type's winning message.

We first establish that a positive mass of types in $\bar{\Omega}_{\mathcal{M}}$ must lose. When $\omega_{\mathcal{M}}^{**} > c_{\mathcal{M}}$, this follows from $F(\bar{\Omega}_{\mathcal{M}}) > F(\omega \geq \omega_{\mathcal{M}}^{**}) = p_{\mathcal{M}}^* + B/\alpha$. When $\omega_{\mathcal{M}}^{**} = c_{\mathcal{M}}$, suppose instead that almost all types in $\bar{\Omega}_{\mathcal{M}}$ win. The payoff bound then implies that almost all types below $c_{\mathcal{M}}$ lose. By continuity of U and monotonicity of L , any message $m < \sup \Omega$ feasible for a type in $\bar{\Omega}_{\mathcal{M}}$ is also feasible for an interval of types immediately below $c_{\mathcal{M}}$. Such a message cannot win, since those losing types could deviate to it. Almost all types in $\bar{\Omega}_{\mathcal{M}}$ must therefore send $\sup \Omega$ and win. This message cannot win under the initial model, since its probability is at least $F(\bar{\Omega}_{\mathcal{M}}) > p_{\mathcal{M}}^*$. Nor can it win through switching. If favorable under the proposed model, its proposed probability is at most B , giving $\lambda(\sup \Omega) \leq B/F(\bar{\Omega}_{\mathcal{M}}) = B/(p_{\mathcal{M}}^* + B/\alpha) < \alpha$, a contradiction.

Let $\omega^\dagger := \inf\{\omega \in \bar{\Omega}_{\mathcal{M}} : \omega \text{ loses}\}$, then every type strictly above $\omega^\dagger$ loses. Suppose for contradiction that $S_1 > 0$. A positive mass of types above ω_0 must then win under the initial model, so $\omega^\dagger > \omega_0$. Now construct $(\hat{\sigma}', \sigma')$ coinciding with $(\hat{\sigma}^*, \sigma^*)$ except that σ' has all types $\omega > \omega^\dagger$ pool on $m' = \max \Omega$. If $\hat{\sigma}^*$ assigns positive probability to m' , move that probability to a message outside Ω under $\hat{\sigma}'$. The message m' was losing under the original profile, since every losing type above $\omega^\dagger$ could send it. Thus the actual and proposed distributions at the original winning messages are unchanged, and those messages remain winning. Under the modified profile, m' is sent almost surely only by types above $\omega^\dagger > \omega_0$ and has zero proposed probability, so it wins under the initial model.

Consequently, almost all types in $\bar{\Omega}_{\mathcal{M}}$ now win. If $\omega_{\mathcal{M}}^{**} > c_{\mathcal{M}}$, this already exceeds the

bound $B/\alpha + p_{\mathcal{M}}^*$. When $\omega_{\mathcal{M}}^{**} = c_{\mathcal{M}}$, some original winning high types sent messages below $\sup \Omega$, since $\sup \Omega$ was losing. By the earlier observation, a positive mass of types below $c_{\mathcal{M}}$ also won and continues to win. The modified winning probability therefore exceeds $F(\bar{\Omega}_{\mathcal{M}}) = B/\alpha + p_{\mathcal{M}}^*$ in this case as well. In either case, we contradict the payoff bound, which applies to every feasible σ' and every proposed model $\hat{\sigma}'$, whether or not they form an equilibrium. Hence $S_1 = 0$.

Sub-case (a). Suppose $B \geq \alpha F(\Omega_{\mathcal{M}}(m^*))$ for some $m^* \in \Omega$. We construct an equilibrium attaining winning probability B/α . Define $\omega_a(m) := \inf\{\omega : U(\omega) \geq m\}$ and $\omega_b(m) := \sup\{\omega : L(\omega) \leq m\}$. Since every type can report truthfully, these boundaries are well-defined and $\Omega_{\mathcal{M}}(m) = [\omega_a(m), \omega_b(m)]$, with infinite endpoints understood as unbounded intervals. By regularity, both boundaries are nondecreasing and continuous, so $F(\Omega_{\mathcal{M}}(m)) = F(\omega_b(m)) - F(\omega_a(m))$ is continuous in m .

Starting from $\{m^*\}$, expand a message interval $[m', m'']$ and let $W := \bigcup_{m \in [m', m'']} \Omega_{\mathcal{M}}(m) = [\omega_a(m'), \omega_b(m'')]$. Its probability increases continuously from $F(\Omega_{\mathcal{M}}(m^*)) \leq B/\alpha$ toward one as the interval expands to cover Ω , since every type can report truthfully. Because $B/\alpha < D_{\mathcal{M}} < 1$, the intermediate value theorem ensures that we can choose $[m', m'']$ with $F(W) = \frac{B}{\alpha}$.

Necessarily $m'' < \sup \Omega$, since otherwise W would contain $\bar{\Omega}_{\mathcal{M}}$, whose probability exceeds B/α . Since $U(\omega_{\mathcal{M}}^{**}) = \sup \Omega > m'$, continuity gives $\omega_a(m') < \omega_{\mathcal{M}}^{**}$. Together with $F(W) = B/\alpha = F([\omega_{\mathcal{M}}^{**}, \omega_{\mathcal{M}}^*])$, this implies $\omega_b(m'') < \omega_{\mathcal{M}}^* \leq \omega_0$.

Now construct σ^* by having each type $\omega \in W$ send $\min\{\max\{\omega, m'\}, m''\}$, a feasible message in $[m', m'']$. Types below $\omega_a(m')$ report truthfully, and types above $\omega_b(m'')$ send $U(\omega)$. These latter two groups send messages outside $[m', m'']$. Their messages below $\sup \Omega$ reveal types below ω_0 . The message $\sup \Omega$ pools types above $\max\{c_{\mathcal{M}}, \omega_b(m'')\} < \omega_{\mathcal{M}}^*$ and must have posterior mean strictly below ω_0 ; otherwise this feasible upper-tail pool would exceed the constrained BP payoff. Thus every message used outside $[m', m'']$ loses under the initial model.

Construct $\hat{\sigma}^*$ as in Part (i). Every type $\omega \geq \omega^*$ randomizes over $[m', m'']$ according to its relative probabilities under σ^* , while every type $\omega < \omega^*$ sends $\hat{m} \in M \setminus \Omega$. Since $P_{\sigma^*}([m', m'']) = B/\alpha$, the induced distribution satisfies $P_{\hat{\sigma}^*}(E) = \alpha P_{\sigma^*}(E \cap [m', m''])$ for every measurable $E \subseteq \Omega$.

First, messages in $[m', m'']$ have $\lambda(m) = \alpha$ almost surely and induce switching. Because all types $\omega \geq \omega^*$ use the same message distribution under $\hat{\sigma}^*$, the receiver's posterior is $F(\cdot | \omega \geq \omega^*)$, with mean ω_0 , so he takes the risky action. Second, actual messages outside

the interval have $\lambda(m) = 0$, so he retains the initial model and takes the safe action. Finally, the proposed-only message $\hat{m}$ induces switching to $F(\cdot|\omega < \omega^*)$ and the safe action. Where likelihood ratios or posteriors are undetermined, choose versions consistent with these actions and the posterior-support restriction.

Every type in W wins, and by definition no type outside W can send a message in $[m', m'']$. Together with the off-path convention, this rules out message and narrative deviations. Since $S_1 = 0$ and $S_2 \leq B/\alpha$ in every equilibrium, the construction attaining $F(W) = B/\alpha$ is sender-optimal.

Sub-case (b). Suppose $B < \alpha F(\Omega_{\mathcal{M}}(m))$ for every $m \in \Omega$. If any type wins by sending $\hat{m}$, every type in $\Omega_{\mathcal{M}}(\hat{m})$ must win; otherwise, it could profitably deviate to $\hat{m}$. The equilibrium winning probability would then be at least $F(\Omega_{\mathcal{M}}(\hat{m})) > B/\alpha$, contradicting $S_1 = 0$ and $S_2 \leq B/\alpha$. Thus all types lose in every equilibrium.

Part (iii). When neither condition for partial manipulation holds, the preceding argument establishes that all types lose in every equilibrium. To establish existence, let σ^* and $\hat{\sigma}^*$ both prescribe $U(\omega)$ for every type. The receiver's action is then independent of which model he adopts. Types below $c_{\mathcal{M}}$ reveal themselves and lose because $c_{\mathcal{M}} \leq \omega_{\mathcal{M}}^{**} < \omega_{\mathcal{M}}^* \leq \omega_0$. Types at or above $c_{\mathcal{M}}$ pool on $\sup \Omega$. This pool's posterior mean must be strictly below ω_0 ; otherwise, it would yield a feasible BP winning probability $F(\omega \geq c_{\mathcal{M}}) > p_{\mathcal{M}}^*$, contradicting **Proposition 1**. Thus every message used under either model induces the safe action. Together with the off-path convention, this rules out message and narrative deviations. $\square$

A.3.6 Proof of **Corollary 5**

Proof. Under either direction of flexibility, every strategy feasible under $\mathcal{M}$ remains feasible under $\mathcal{M}'$. Thus $p_{\mathcal{M}'}^* \geq p_{\mathcal{M}}^*$, $D_{\mathcal{M}'} \leq D_{\mathcal{M}}$, and $\omega_{\mathcal{M}'}^* \leq \omega_{\mathcal{M}}^*$. For comparisons with narratives, if $B \geq \alpha D_{\mathcal{M}'}$, the sender obtains payoff 1 under $\mathcal{M}'$ and cannot be worse off. In Cases (i)(b) and (ii)(b), we therefore need only consider $B < \alpha D_{\mathcal{M}'} \leq \alpha D_{\mathcal{M}}$. Both partial-manipulation cutoffs are then defined, with $\omega_{\mathcal{M}'}^{**} \leq \omega_{\mathcal{M}}^{**}$.

Case (i)(a). Downward flexibility leaves U and hence the set of credible types unchanged. Since $\omega_{\mathcal{M}'}^* \leq \omega_{\mathcal{M}}^*$, we have $\tilde{\omega}_{\mathcal{M}'}^* \leq \tilde{\omega}_{\mathcal{M}}^*$ whenever the latter exists. By **Proposition 2**, the sender's payoff weakly increases. If $\tilde{\omega}_{\mathcal{M}}^*$ does not exist, her original payoff is zero, so she cannot be worse off.

Case (i)(b). Downward flexibility also leaves $c_{\mathcal{M}'} = c_{\mathcal{M}}$. If $\omega_{\mathcal{M}}^{**} < c_{\mathcal{M}}$, then $\omega_{\mathcal{M}'}^{**} \leq \omega_{\mathcal{M}}^{**} < c_{\mathcal{M}'}$, so **Theorem 3** implies that the sender's payoff weakly increases from $p_{\mathcal{M}}^* + B/\alpha$ to $p_{\mathcal{M}'}^* + B/\alpha$. If $\omega_{\mathcal{M}}^{**} \geq c_{\mathcal{M}}$, her original payoff is B/α when $\mathcal{M}$ satisfies the feasible-pool

condition and zero otherwise. A strict loss therefore requires that condition to hold under $\mathcal{M}$. Under $\mathcal{M}'$, she obtains $p_{\mathcal{M}'}^* + B/\alpha$ if $\omega_{\mathcal{M}'}^{**} < c_{\mathcal{M}'}$; otherwise, she obtains B/α or zero according to whether the feasible-pool condition holds. Thus she is strictly worse off exactly when both cutoffs are weakly above their credibility boundaries and the feasible-pool condition holds under $\mathcal{M}$ but fails under $\mathcal{M}'$, together with $B < \alpha D_{\mathcal{M}'}$.

Case (ii)(a). Since upward flexibility raises U , we have $c_{\mathcal{M}'} \leq c_{\mathcal{M}}$. For regular technologies, [Proposition 2](#) implies that the sender's payoff without narratives is $p_{\mathcal{M}}^*$ if $\omega_{\mathcal{M}}^* \leq c_{\mathcal{M}}$ and zero otherwise, and similarly under $\mathcal{M}'$. Because $p_{\mathcal{M}'}^* \geq p_{\mathcal{M}}^*$, a strict loss occurs exactly when the original payoff is positive and the new payoff is zero, i.e. $c_{\mathcal{M}'} < \omega_{\mathcal{M}'}^* \leq \omega_{\mathcal{M}}^* \leq c_{\mathcal{M}}$.

Case (ii)(b). If $\omega_{\mathcal{M}'}^{**} < c_{\mathcal{M}'}$, the sender obtains $p_{\mathcal{M}'}^* + B/\alpha$. This weakly exceeds $p_{\mathcal{M}}^* + B/\alpha$, the payoff bound under $\mathcal{M}$, so she cannot be worse off. Suppose instead that $\omega_{\mathcal{M}'}^{**} \geq c_{\mathcal{M}'}$. Her new payoff is B/α or zero, depending on whether $\mathcal{M}'$ satisfies the feasible-pool condition. If $\omega_{\mathcal{M}}^{**} < c_{\mathcal{M}}$, her original payoff is $p_{\mathcal{M}}^* + B/\alpha > B/\alpha$, so she is strictly worse off. This is precisely the case $c_{\mathcal{M}'} \leq \omega_{\mathcal{M}'}^{**} \leq \omega_{\mathcal{M}}^{**} < c_{\mathcal{M}}$. Finally, if $\omega_{\mathcal{M}}^{**} \geq c_{\mathcal{M}}$ as well, the payoff under either technology is B/α when its feasible-pool condition holds and zero otherwise. A strict loss therefore occurs exactly when that condition holds under $\mathcal{M}$ but fails under $\mathcal{M}'$. Together with $B < \alpha D_{\mathcal{M}'}$, these are the stated conditions. $\square$

A.3.7 Proof of [Theorem 4](#)

Proof. Sufficiency. Suppose conditions (i) and (ii) hold. By condition (i) and monotonicity, $L(\omega) \leq U(x) \leq U(\omega)$ whenever $\omega \geq x$. Thus, for any cutoff $x \in \Omega$, all types above it can pool on $U(x)$. By condition (i), all types $\omega \geq \omega^*$ can pool on $U(\omega^*)$, while lower types report truthfully, so $p_{\mathcal{M}}^* = p^*$. If $B \geq \alpha D$, [Theorem 2](#) gives payoff 1.

Now suppose $B < \alpha D$, and define ω^{**} by $F(\omega \geq \omega^{**}) = p^* + p^*/\alpha$. Under σ^* , let types $\omega \geq \omega^*$ send $U(\omega^*)$, types $\omega \in [\omega^{**}, \omega^*)$ send $U(\omega^{**})$, and lower types send $L(\omega)$. Under $\hat{\sigma}^*$, let types $\omega \geq \omega^*$ send $U(\omega^{**})$ and lower types send $L(\omega)$. The preceding observation ensures feasibility of σ^* , while condition (ii) gives $U(\omega^{**}) < U(\omega^*)$. Moreover, choosing any $x < \omega^{**}$, conditions (i) and (ii) imply $L(\omega) \leq U(x) < U(\omega^{**})$ for every ω , so neither winning message coincides with any remaining message.

By construction, $\lambda(U(\omega^{**})) = B/(B/\alpha) = \alpha$, so the receiver adopts $\hat{\sigma}^*$ and takes the risky action, since $\mathbb{E}[\omega | \omega \geq \omega^*] = \omega_0$. Following $U(\omega^*)$, he retains σ^* and takes the same action. All remaining on-path messages induce the safe action under either model. Condition (ii) prevents types below ω^{**} from accessing either winning message. Together with the off-path convention in [Appendix A.1](#), this rules out message and narrative

deviations. The equilibrium therefore attains $p^* + p^*/\alpha$.

Necessity of (i). Suppose condition (i) fails, so there exist $\omega'' < \omega'$ with $U(\omega'') < L(\omega')$. Right-continuity of U and left-continuity of L let us choose both types in the interior of Ω . It suffices to construct one admissible environment wherein $\mathcal{M}$ fails to attain the payoff bound.

Set $\alpha = 1$ and $\omega_0 = \omega'$, and choose $a \in (\inf \Omega, \omega'')$. Take distributions G and H with finite first moments and continuous, strictly positive densities g and h on $\Omega \cap [a, \infty)$ and $\Omega \cap (-\infty, a]$, respectively, satisfying $\mathbb{E}_G[\omega] = \omega_0$ and $h(a) = g(a)/2$. Define $F = \frac{2}{3}H + \frac{1}{3}G$. The matching condition ensures that F admits a continuous, strictly positive density on Ω . Moreover, $\mathbb{E}_F[\omega] < \omega_0$ and $F(\omega \geq \omega_0) > 0$, so the environment is admissible.

By construction, $\omega^* = a$ and $p^* = 1/3$. Types in $[a, \omega'']$ cannot share a message with any type at or above ω_0 , since $U(\omega) \leq U(\omega'') < L(\omega_0)$. They therefore cannot win under a feasible BP strategy. By [Proposition 1](#), $\omega_{\mathcal{M}}^* \geq \omega'' > a = \omega^*$, so $p_{\mathcal{M}}^* < p^*$. The payoff bound from the proof of [Theorem 2](#) then implies that every equilibrium payoff is at most $p_{\mathcal{M}}^* + p^* < 2p^* = 2/3$. Thus $\mathcal{M}$ is not maximally persuasive.

Necessity of (ii). Suppose condition (ii) fails. Since U is nondecreasing, there exist interior types $\omega'' < \omega'$ such that U is constant on $[\omega'', \omega']$. Set $\alpha = 1$, $\omega_0 = \omega'$, and choose $\hat{\omega} \in (\omega'', \omega')$.

Take distributions G , H , and J with finite first moments and continuous, strictly positive densities g , h , and j on $\Omega \cap [\hat{\omega}, \infty)$, $[\omega'', \hat{\omega}]$, and $\Omega \cap (-\infty, \omega'']$, respectively, satisfying $\mathbb{E}_G[\omega] = \omega_0$, $h(\hat{\omega}) = g(\hat{\omega})/2$, and $j(\omega'') = 2h(\omega'')$. Define $F = \frac{1}{4}J + \frac{1}{2}H + \frac{1}{4}G$. The matching conditions ensure that F admits a continuous, strictly positive density on Ω . Since $\mathbb{E}_F[\omega] < \omega_0$ and $F(\omega \geq \omega_0) > 0$, the environment is admissible.

By construction, $\omega^* = \hat{\omega}$ and $p^* = 1/4$. Suppose an equilibrium attains the payoff bound $2p^* = 1/2$. Let S_1 and S_2 denote the probabilities of inducing the risky action without and with switching, respectively. Since each is bounded above by $p^* = 1/4$, both bounds must bind. Equality $S_1 = p^*$ implies that the actual messaging strategy attains the unconstrained BP bound. By [Lemma 1](#), almost all types above $\hat{\omega}$ therefore win without switching, while almost no type below $\hat{\omega}$ does.

For any $x \in [\omega'', \hat{\omega})$ and $y \in (\hat{\omega}, \omega')$, monotonicity of L and constancy of U imply $M(y) \subseteq M(x)$. Each such lower type can thus imitate a winning higher type and must also win in equilibrium. Almost all these lower types must win through switching, so $S_2 \geq F([\omega'', \hat{\omega})) = 1/2 > 1/4$, contradicting the switching bound. Thus $\mathcal{M}$ is not maximally persuasive. $\square$